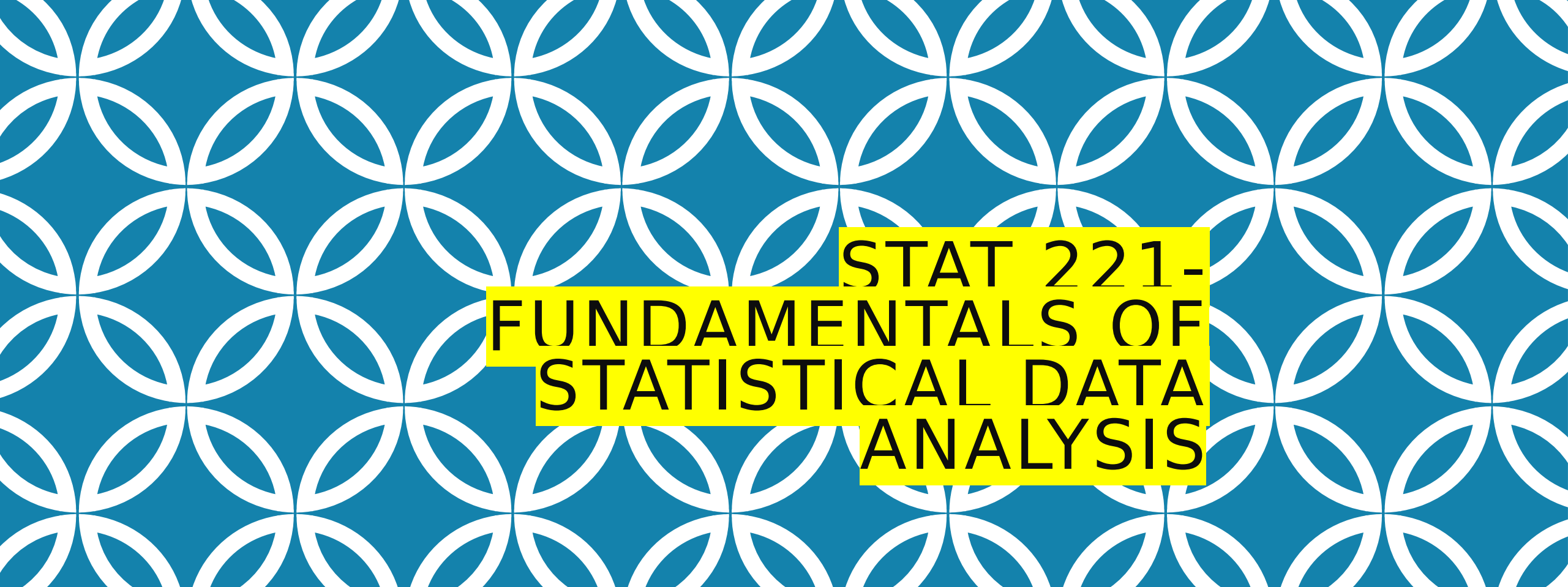
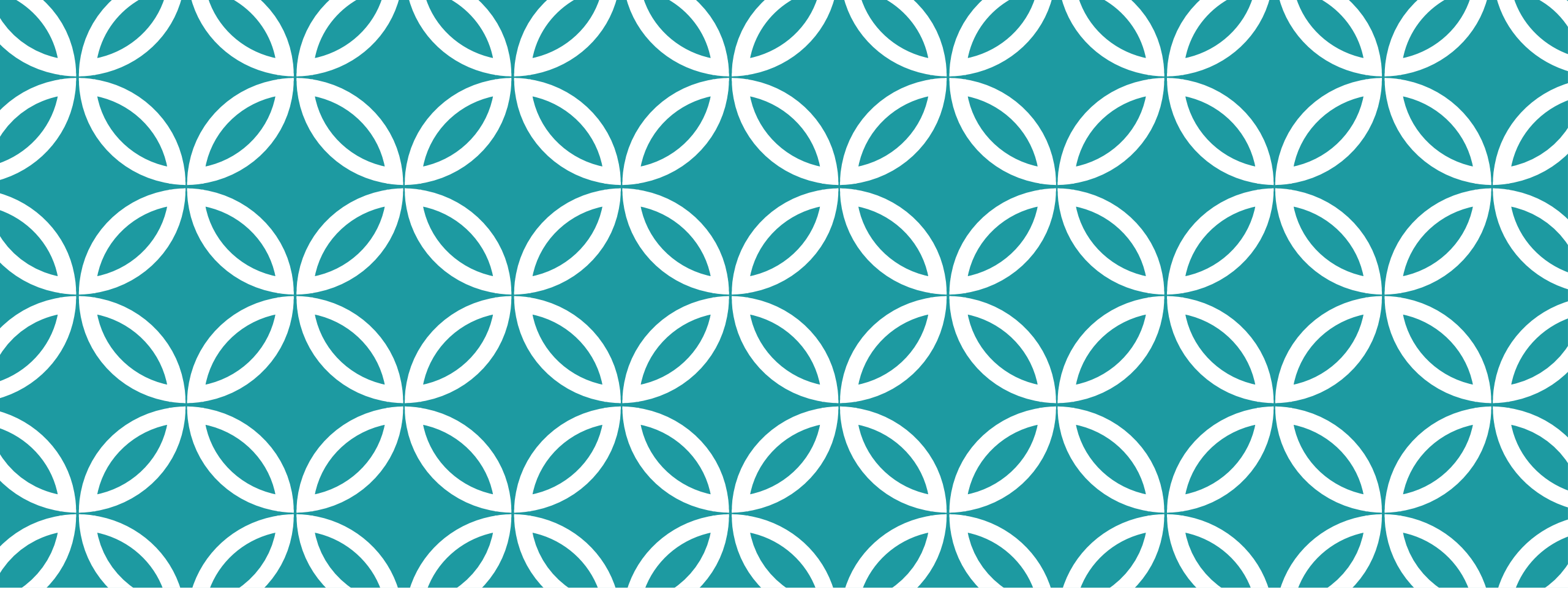




STAT 221- FUNDAMENTALS OF STATISTICAL DATA ANALYSIS

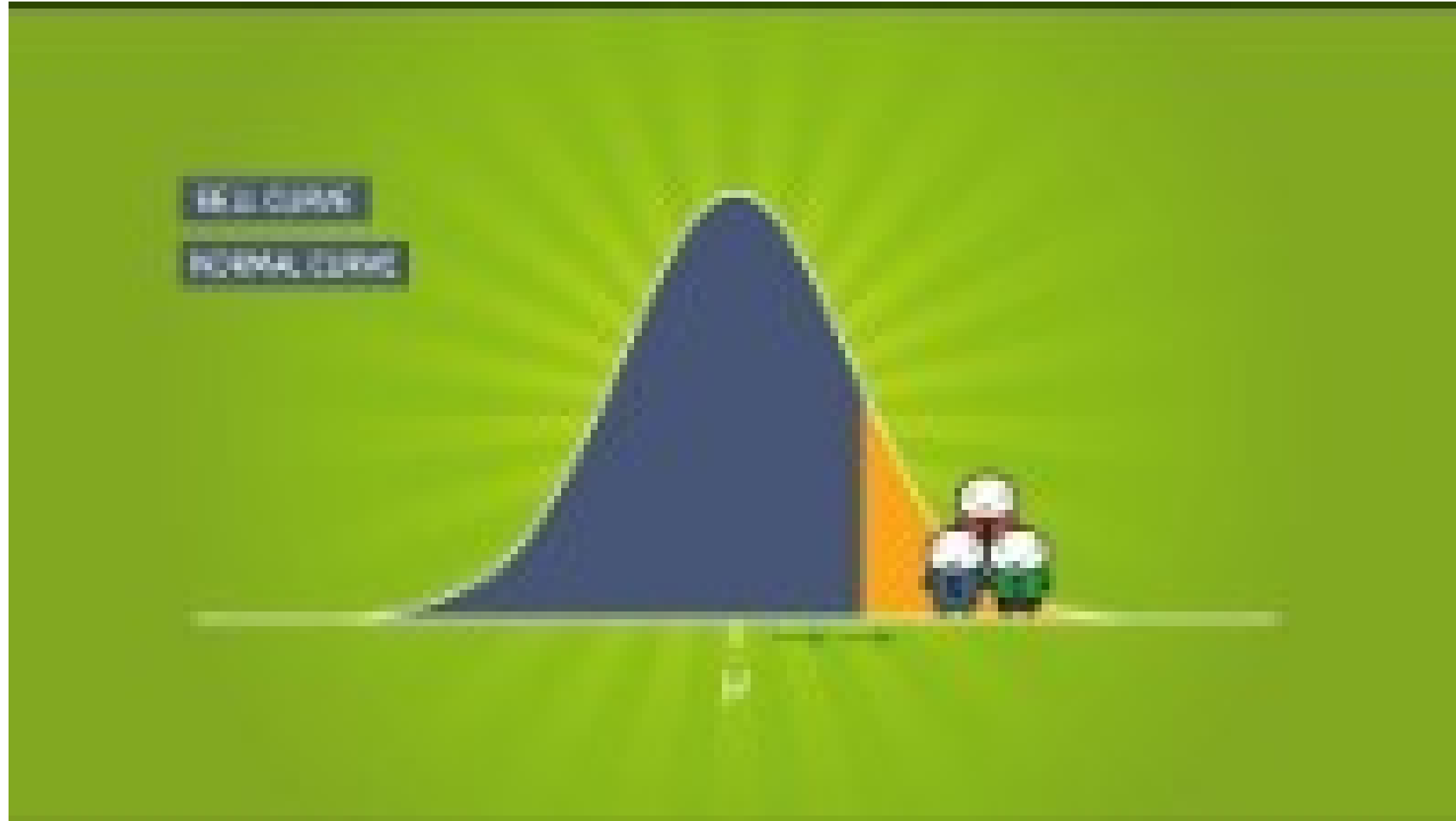
Week-11: Normal Distribution, Central
Limit Theorem

Shivakumar Jolad



NORMAL DISTRIBUTION |

NORMAL DISTRIBUTION



<https://www.youtube.com/watch?v=mtbJbDwqWLE>

#5

**NORMAL
DISTRIBUTION**

NORMAL DISTRIBUTIONS

This *pdf* is the most popular distribution for continuous random variables

First described de Moivre in 1733

Elaborated in 1812 by Laplace

Describes some natural phenomena

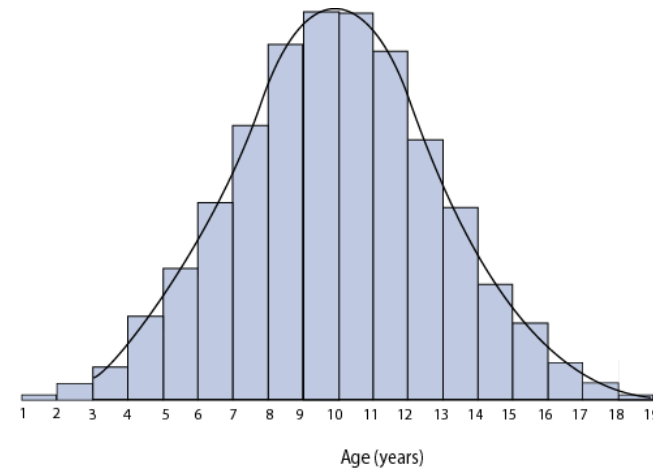
More importantly, describes sampling characteristics of totals and means

NORMAL PROBABILITY DENSITY FUNCTION

Recall: continuous random variables are described with probability density function (***pdfs***) curves

Normal *pdfs* are recognized by their typical bell-shape

Figure: Age distribution of a pediatric population with overlying Normal *pdf*

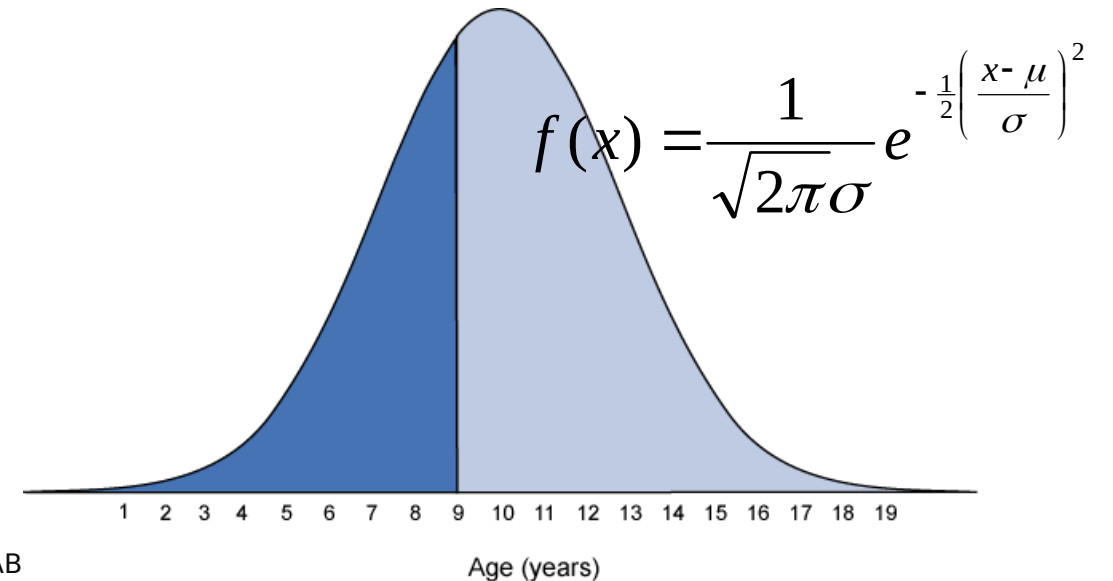
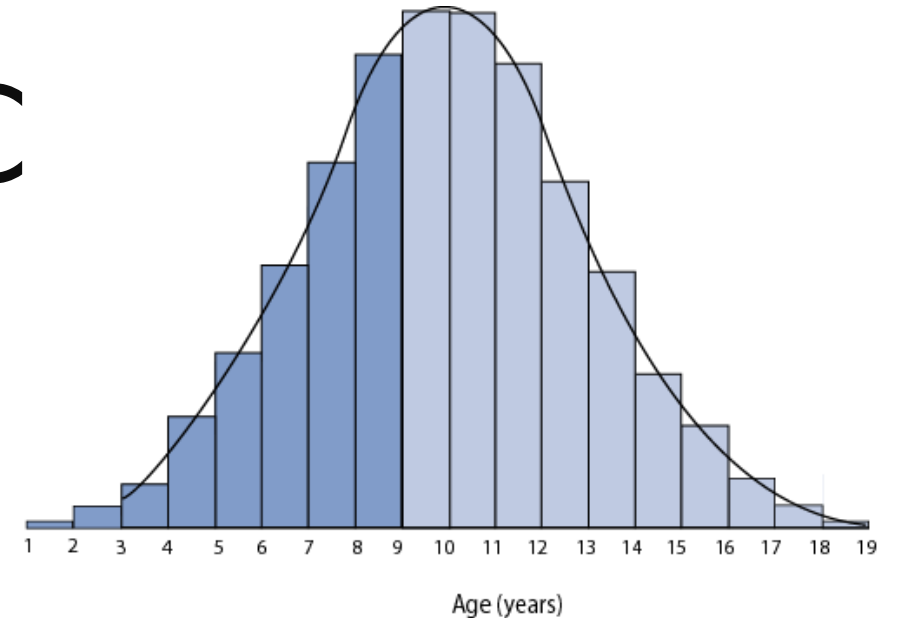


AREA UNDER THE C

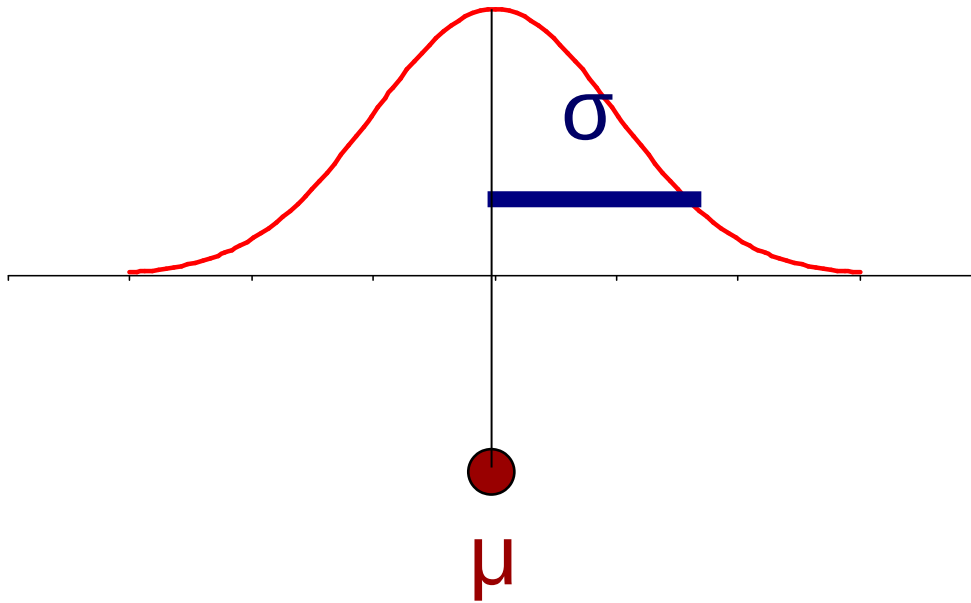
pdfs should be viewed almost like a histogram

Top Figure: The **darker bars** of the histogram correspond to ages ≤ 9 (~40% of distribution)

Bottom Figure: **shaded area under the curve (AUC)** corresponds to ages ≤ 9 (~40% of area)



MEAN AND STANDARD DEVIATION OF NORMAL DENSITY



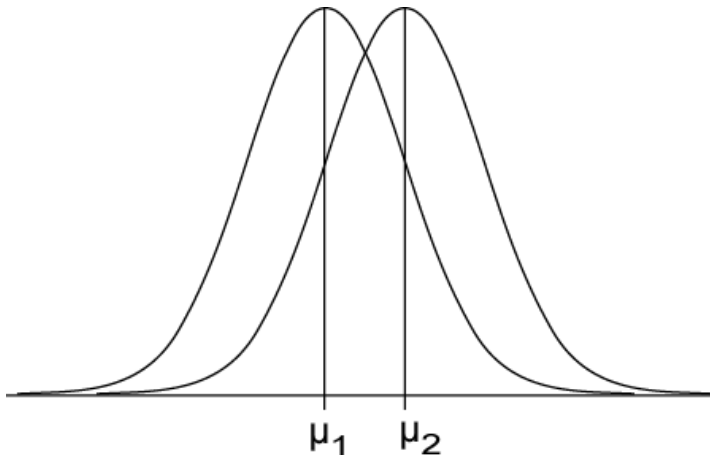
PARAMETERS *MU* AND *SIGMA*

Normal *pdfs* have two **parameters**

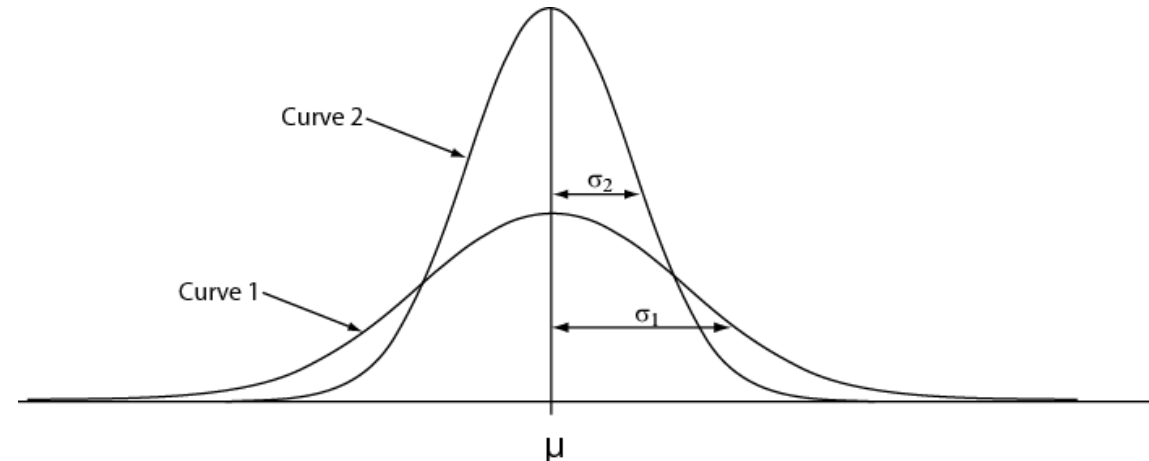
μ - expected value (mean “mu”)

σ - standard deviation (sigma)

μ controls location



σ controls spread



STANDARD DEVIATION

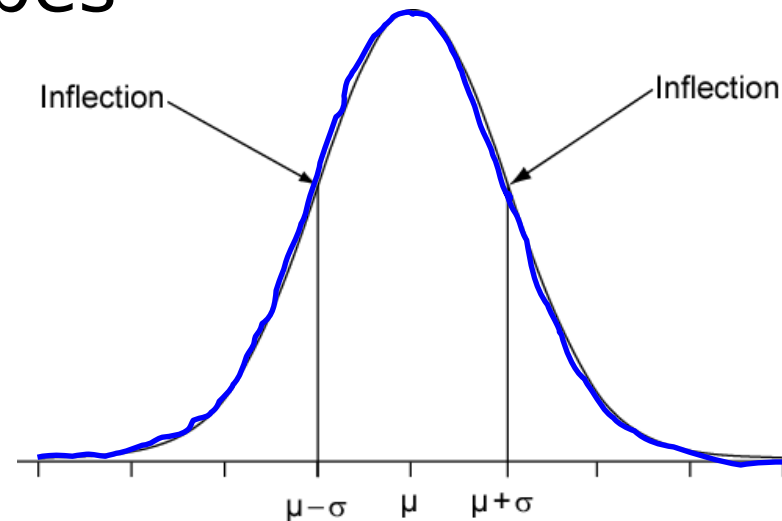
σ

Points of inflections one σ below and above μ

Practice **sketching** Normal curves

Feel inflection points (where slopes change)

Label horizontal axis with σ landmarks



TWO TYPES OF MEANS AND STANDARD DEVIATIONS

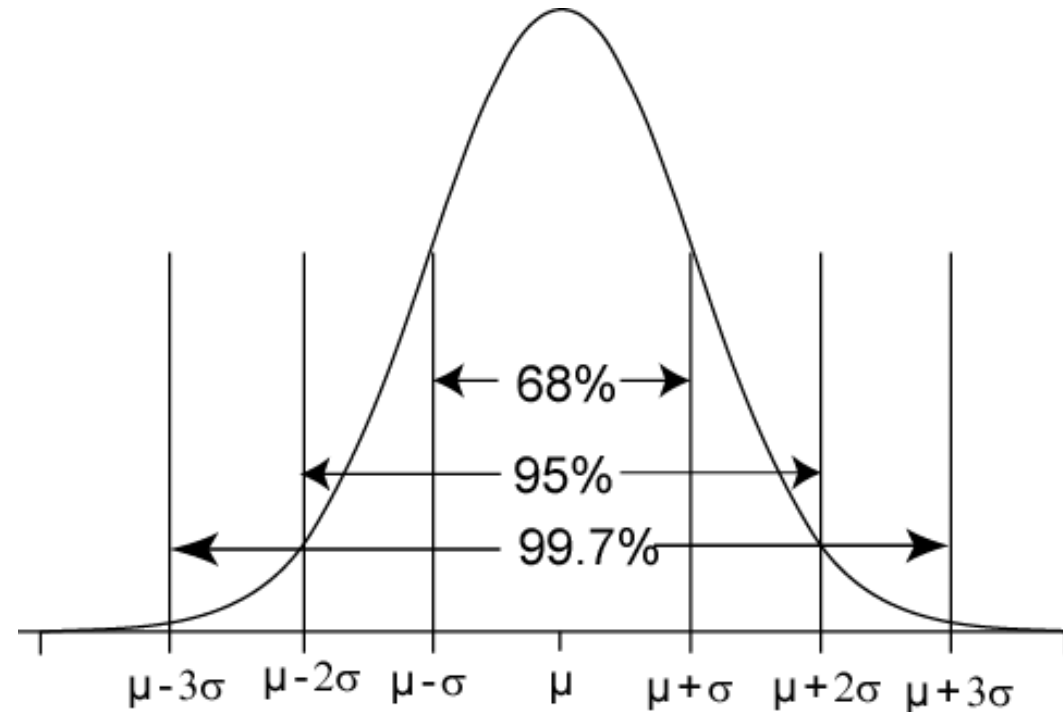
- The mean and standard deviation from the *pdf* (denoted μ and σ) are ***parameters***
- The mean and standard deviation from a sample ($\bar{x}$ -“xbar” and s) are ***statistics***
- Statistics and parameters are related, but are not the same thing!

68-95-99.7 RULE FOR NORMAL DISTRIBUTIONS

68% of the AUC within $\pm 1\sigma$ of μ

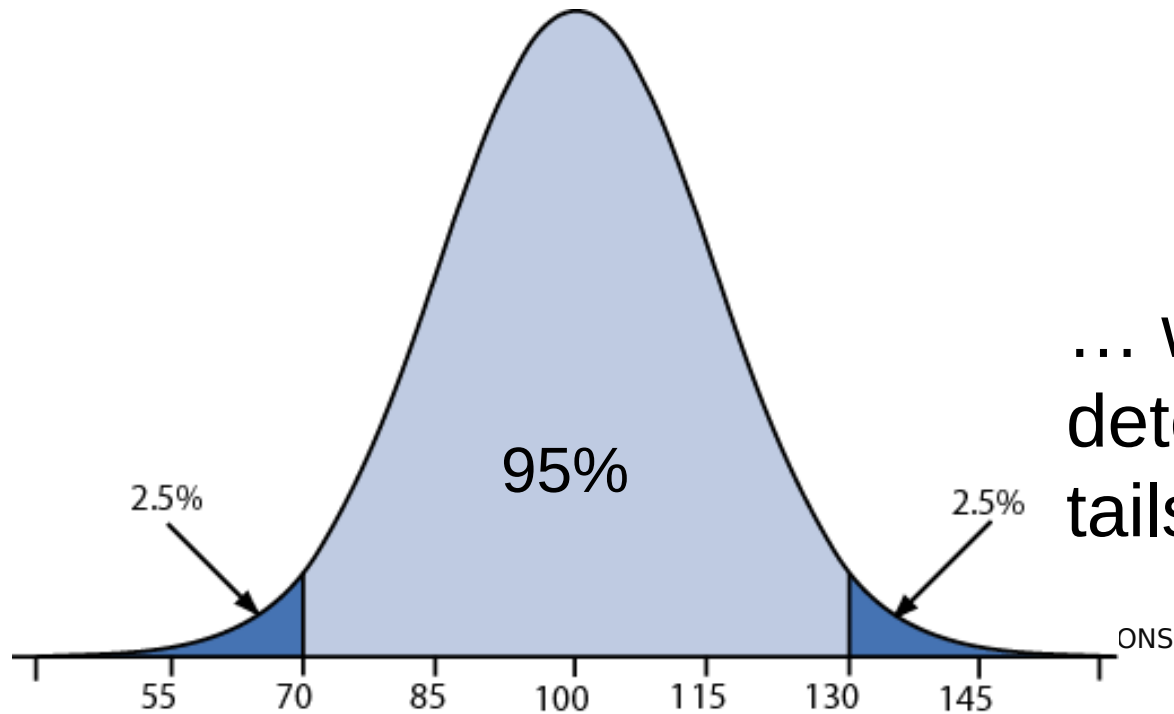
95% of the AUC within $\pm 2\sigma$ of μ

99.7% of the AUC within $\pm 3\sigma$ of μ



SYMMETRY IN THE TAILS

Because the Normal curve is symmetrical and the total AUC is exactly 1...



... we can easily determine the AUC in tails

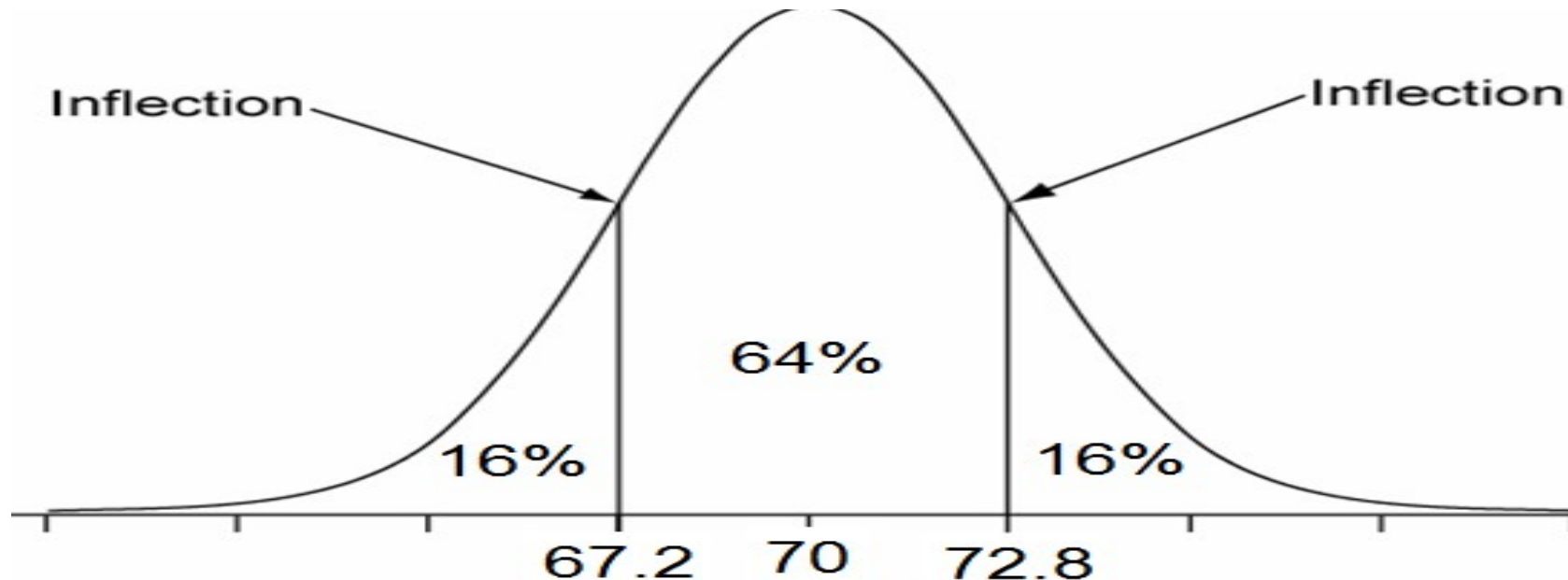
EXAMPLE: MALE HEIGHT

Male height: Normal with $\mu = 70.0''$ and $\sigma = 2.8''$

68% within $\mu \pm \sigma = 70.0 \pm 2.8 = 67.2$ to 72.8

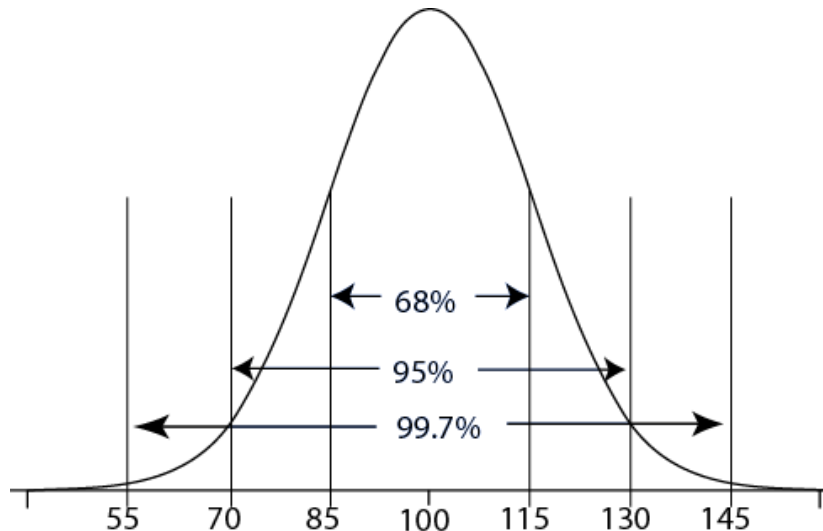
32% in tails (below $67.2''$ and above $72.8''$)

16% below $67.2''$ and 16% above $72.8''$ (symmetry)



EXAMPLE: 68-95-99.7 RULE

Wechsler adult intelligence scores: Normally distributed with $\mu = 100$ and $\sigma = 15$; $X \sim N(100, 15)$



68% of scores within
 $\mu \pm \sigma$
 $= 100 \pm 15$
 $= 85 \text{ to } 115$

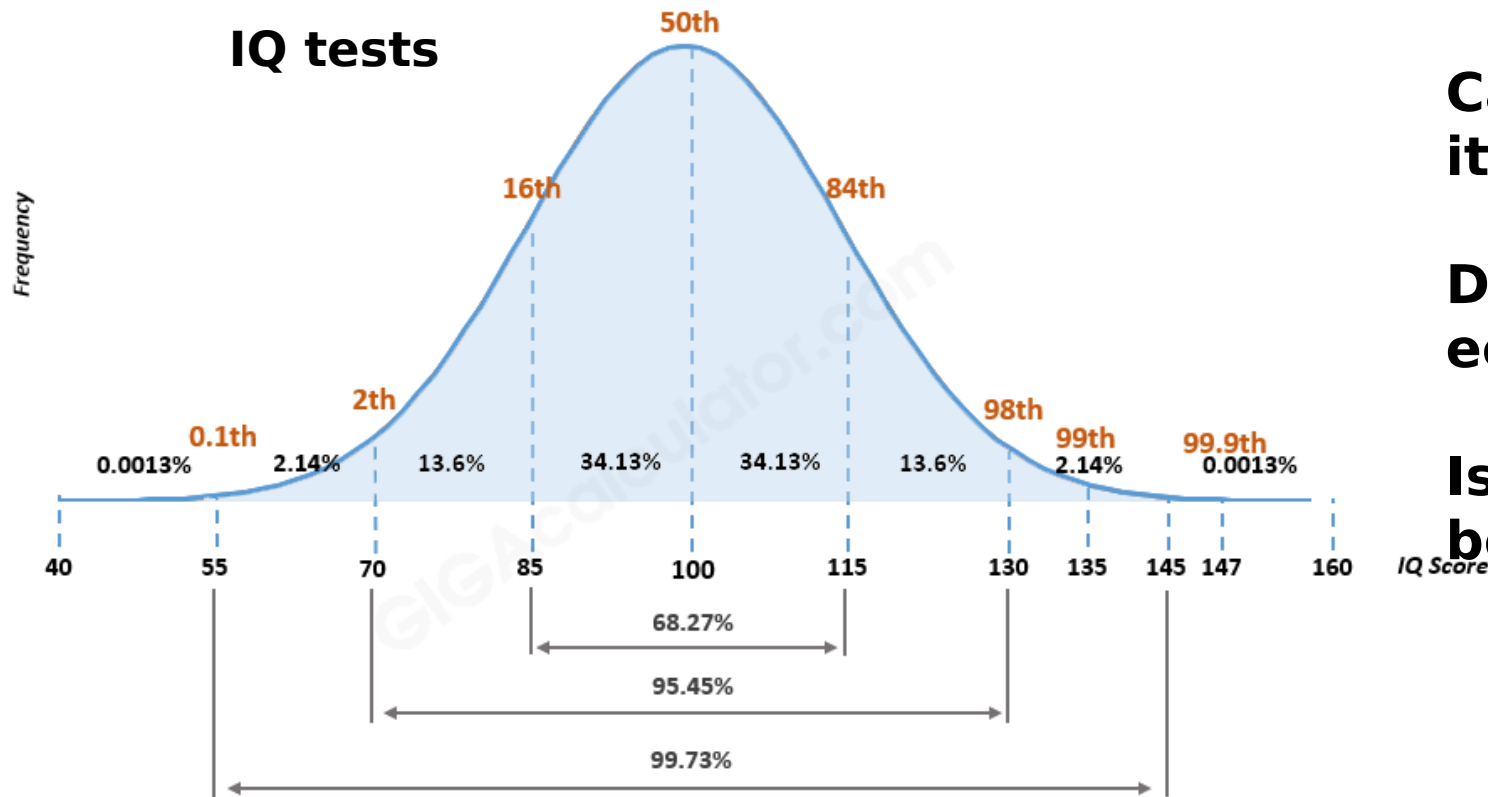
95% of scores within
 $\mu \pm 2\sigma$
 $= 100 \pm (2)(15)$
 $= 70 \text{ to } 130$

99.7% of scores in
 $\mu \pm 3\sigma =$
 $100 \pm (3)(15)$
 $= 55 \text{ to } 145$

INTELLIGENCE, LIFE CHANCES, AND INEQUALITY

IQ Scale with percentiles

IQ tests



What constitutes intelligence?

Can standardized tests measure it?

Does it vary across social and economic groups?

Is IQ natural or acquired? Or both? Role of Nature and Nurture

German sociologist Max Weber (1864–1920) theorized that “life chances are probabilistic” and that being wealthy does not guarantee pathways to upward mobility, it only increases the likelihood that those pathways will present themselves.

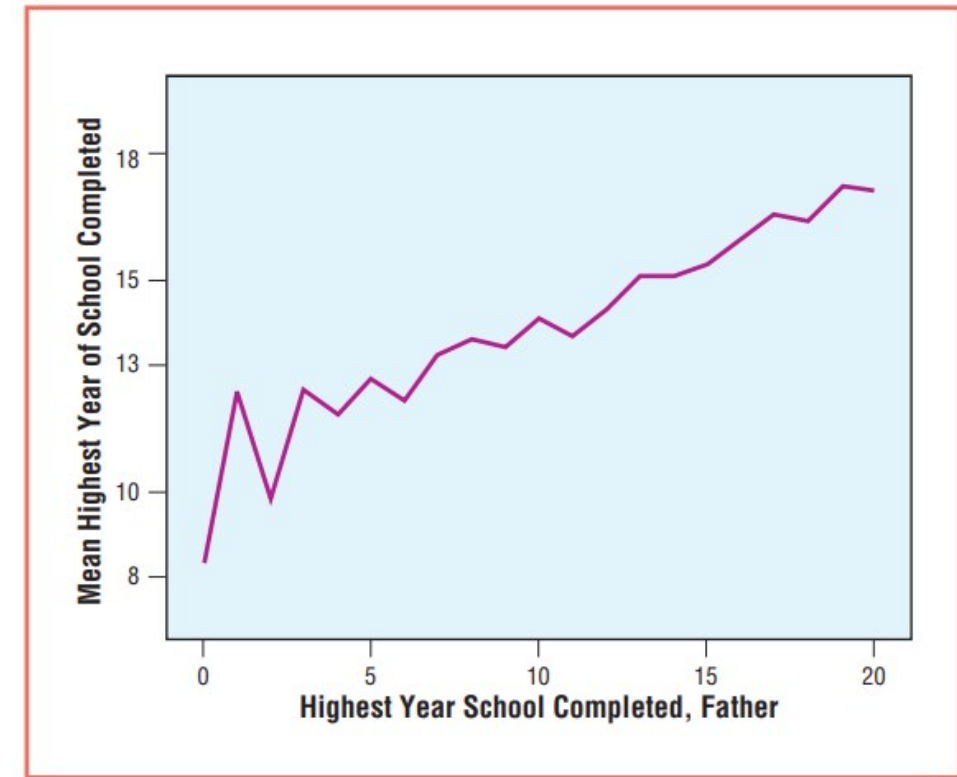


Figure 1 Mean Years of School Completed by Father's Education

REEXPRESSION OF NON-NORMAL RANDOM VARIABLES

Many variables are *not* Normal but can be re-expressed with a *mathematical transformation* to be Normal

Example of mathematical transforms used for this purpose:

- logarithmic
- exponential
- square roots

Review logarithmic transformations...

LOGARITHMS

Logarithms are exponents of their base

Common log
(base 10)

□ $\log(10^0) = 0$

□ $\log(10^1) = 1$

□ $\log(10^2) = 2$

Natural ln (base e)

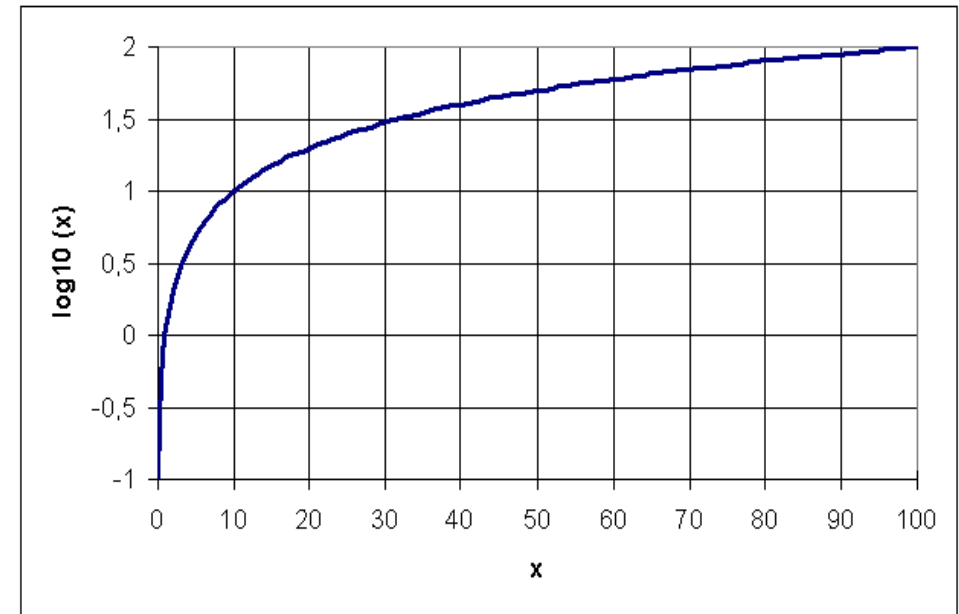
□ $\ln(e^0) = 0$

□ $\ln(e^1) = 1$

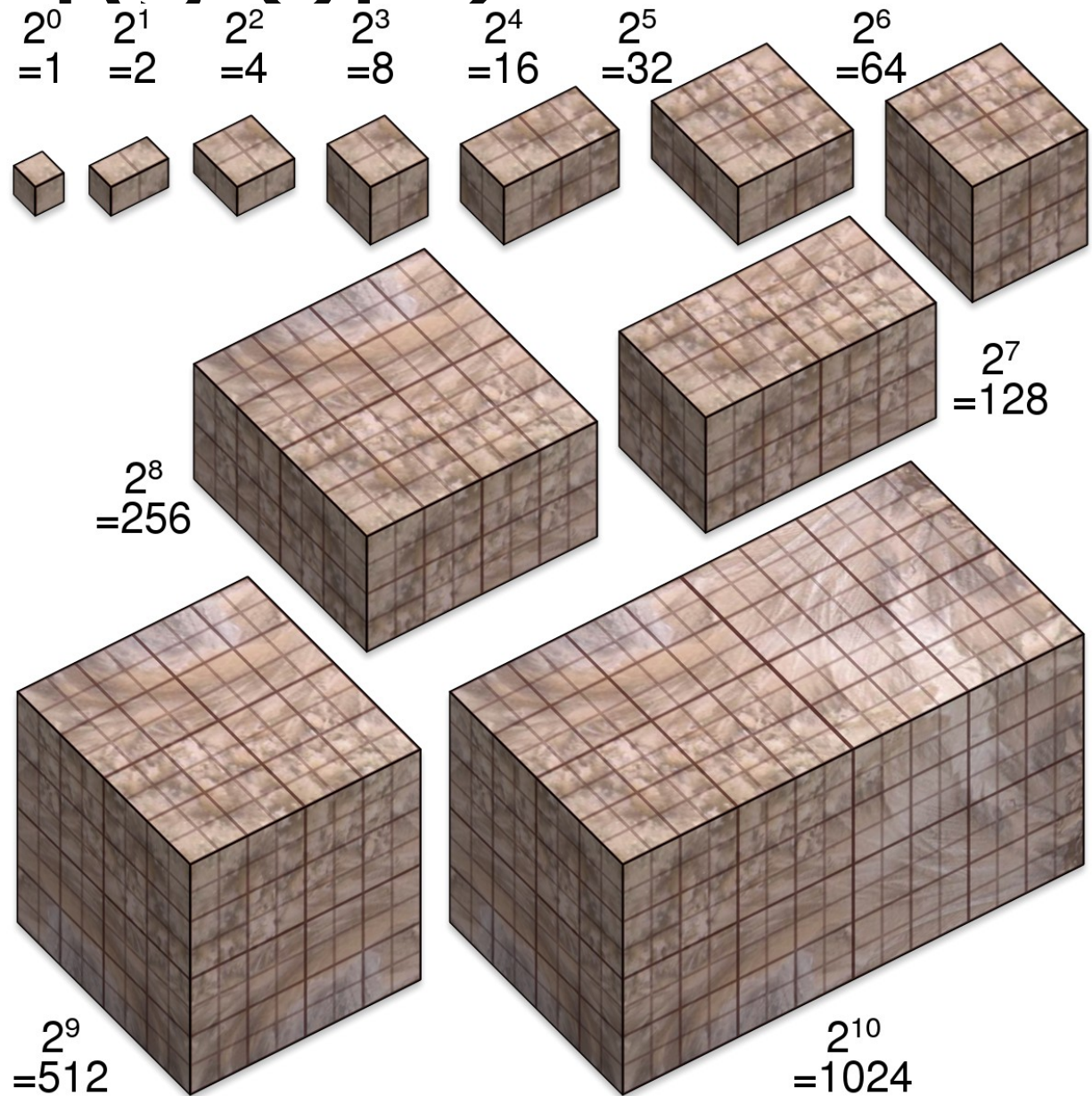
□ $\ln(e^2) = 2$

□ ...

Base 10 log function



LOG IN POWERS OF 2



2 $\log_2 2 = 1$

4 $\log_2 2^2 = 2$

8 $\log_2 2^3 = 3$

1024 $\log_2 2^{10} = 10$

1,048,576 $\log_2 2^{20} = 20$

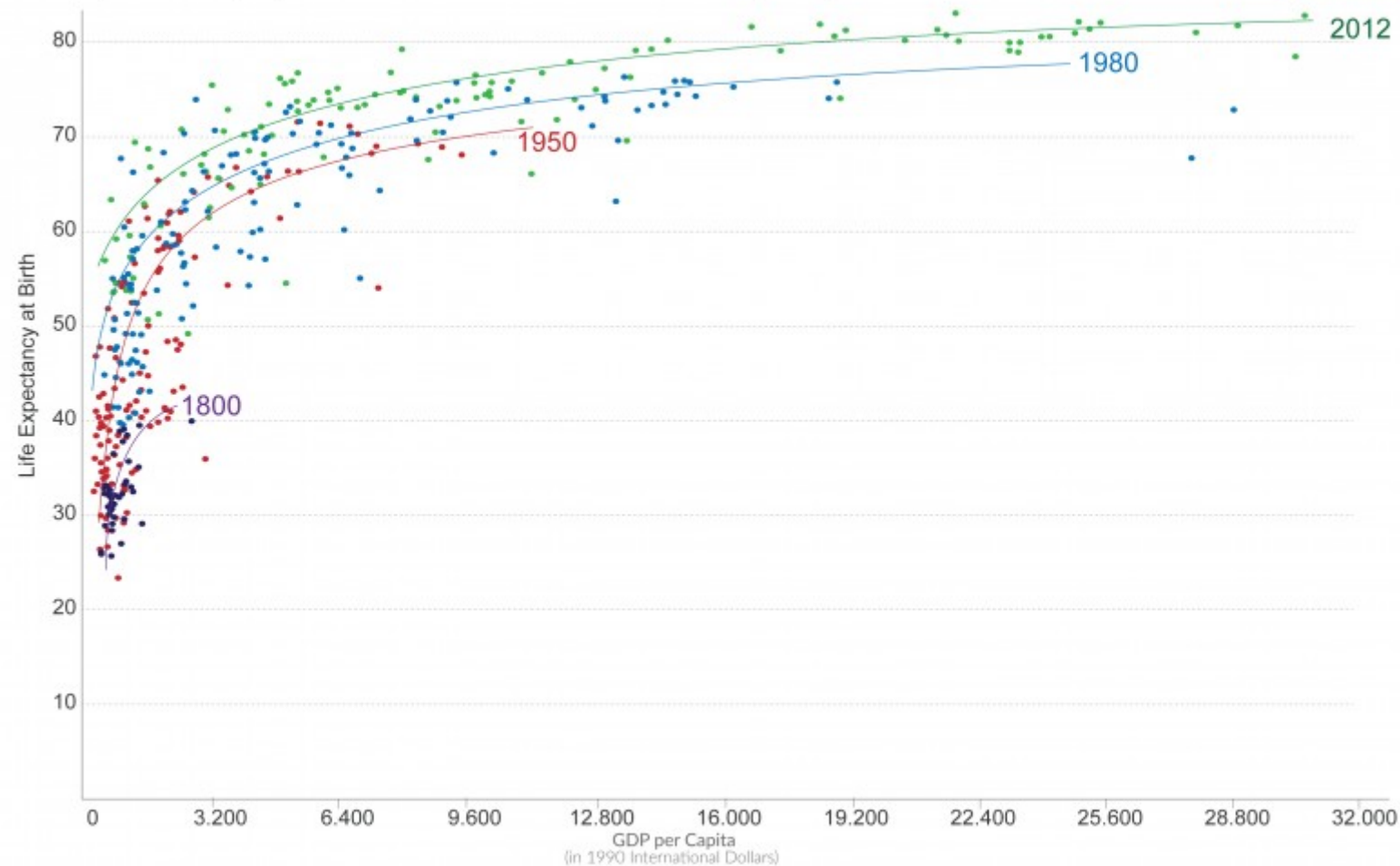
1,099,511,627,776 $\log_2 2^{40} = 40$

INCOME SCALE

Life Expectancy vs. GDP per capita in 1800, 1950, 1980 and 2012

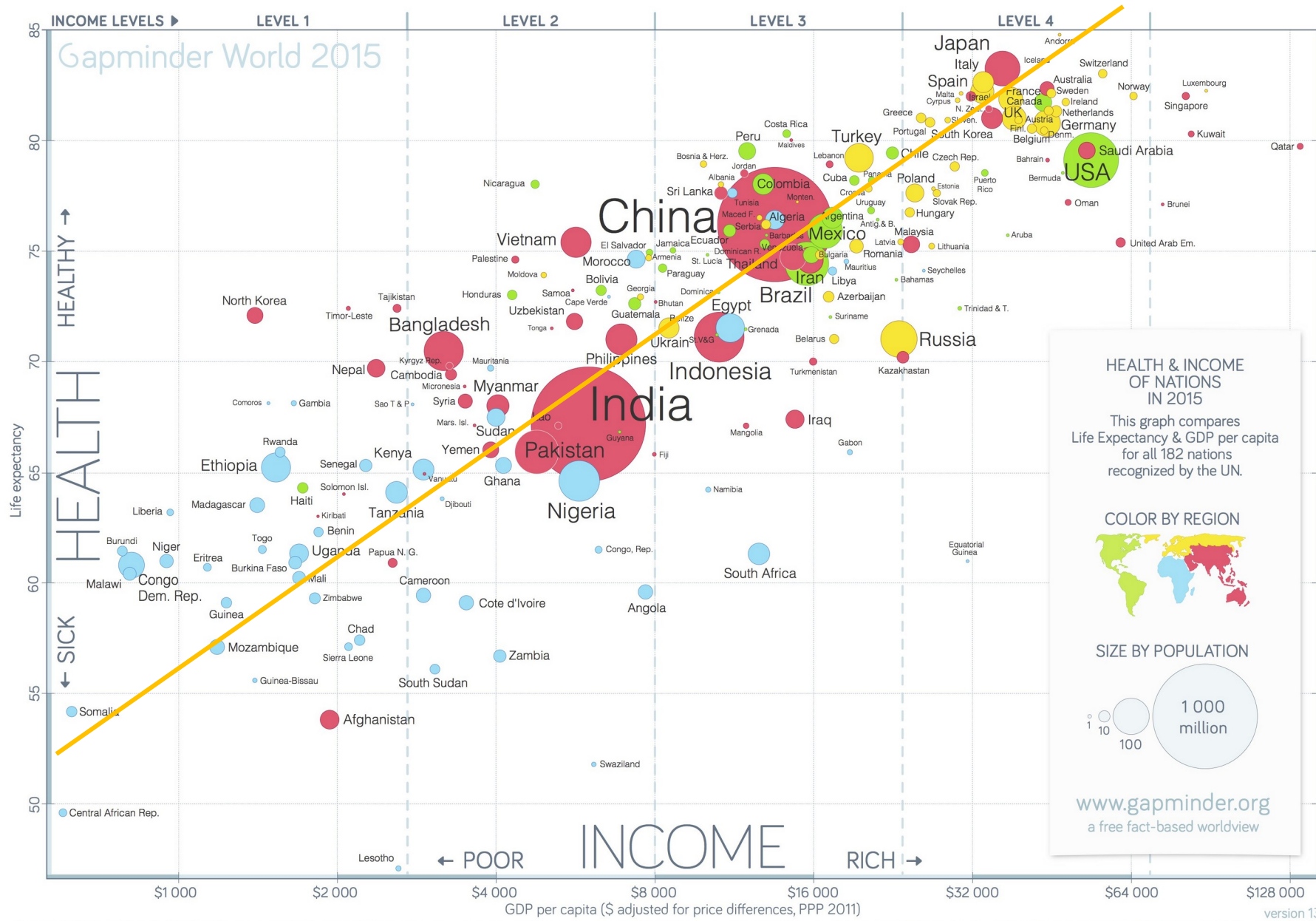
To allow comparisons of GDP per capita over time and between different countries the measure is adjusted for price differences between countries and inflation.

Our World
in Data



Data sources: Data on life expectancy are from Gapminder.org; data on GDP per capita are from the 'New Maddison Project Database'.
OurWorldinData.org - Research and data to make progress against the world's largest problems.

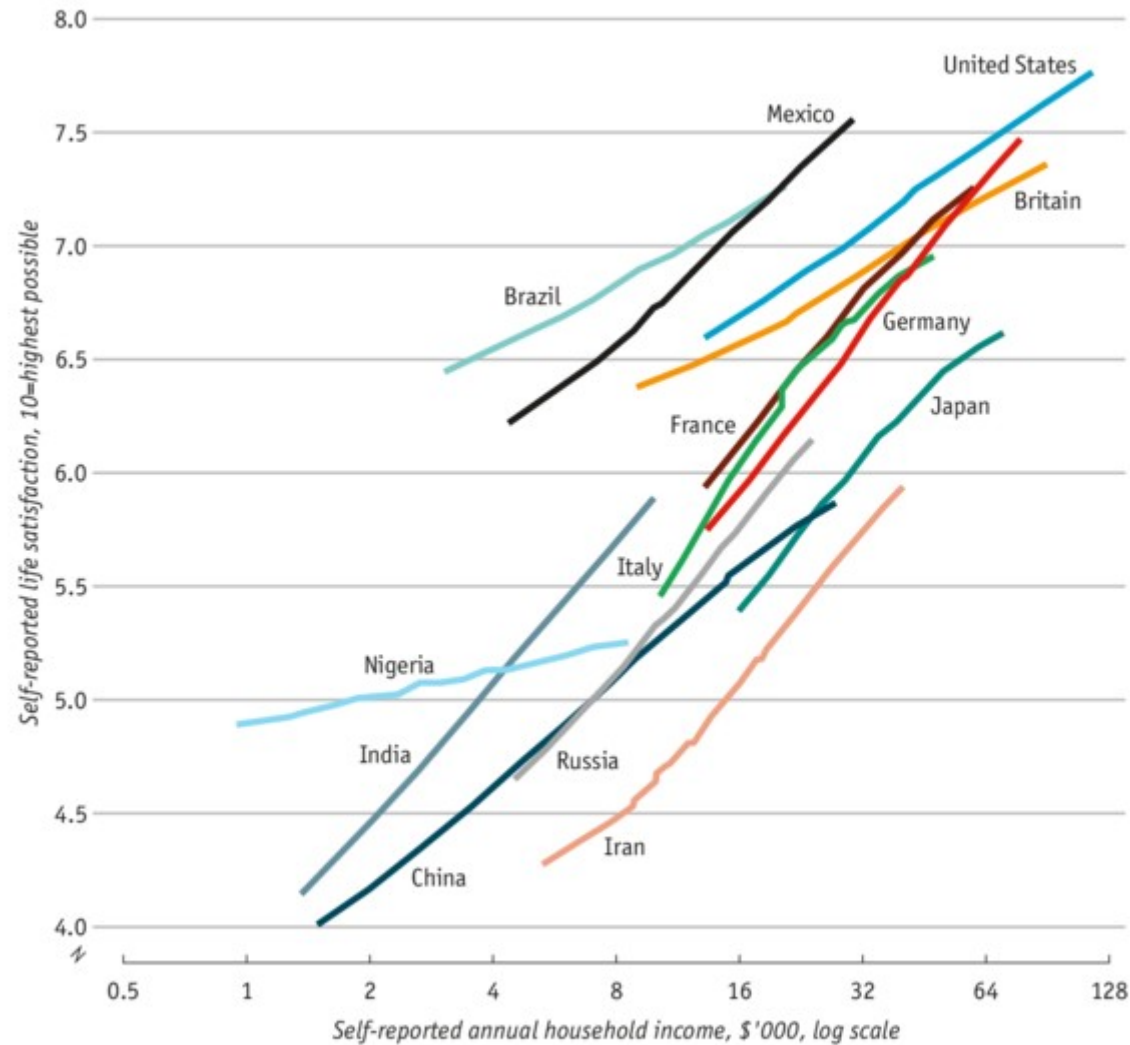
Licensed under CC-BY by the author Max Roser



LINEAR-LOG VARIATION

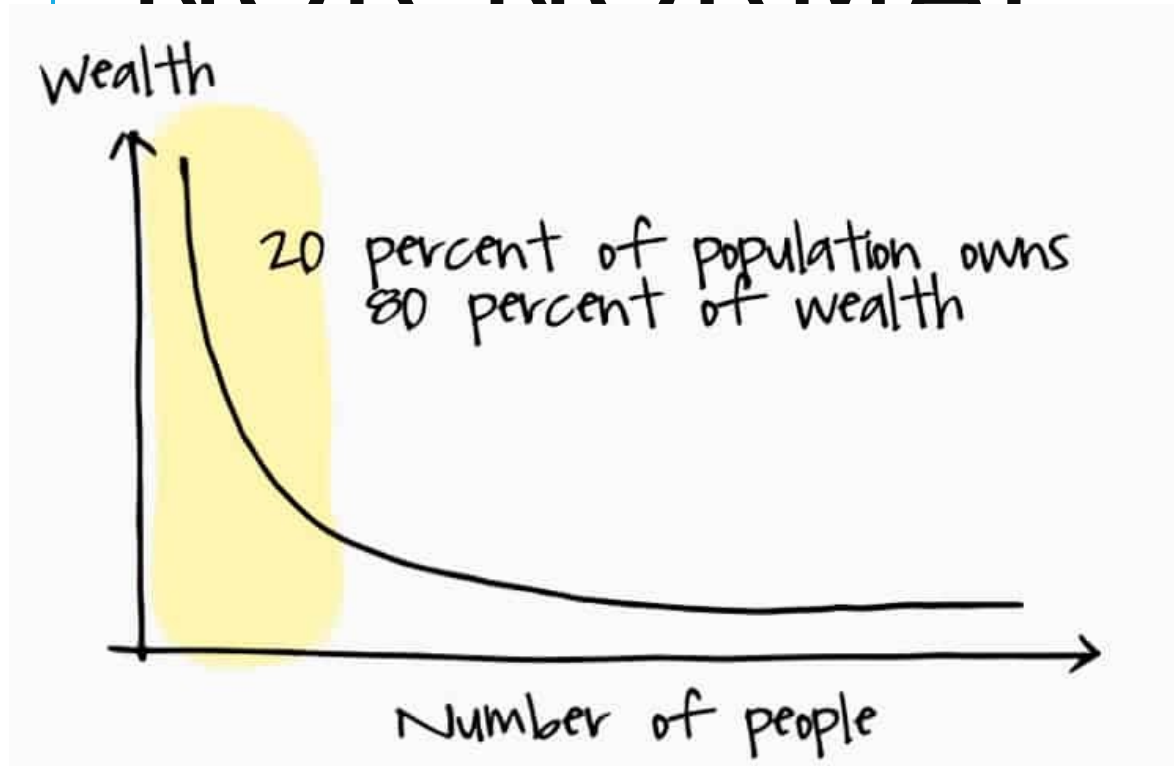
Life satisfaction and income

2012 or latest

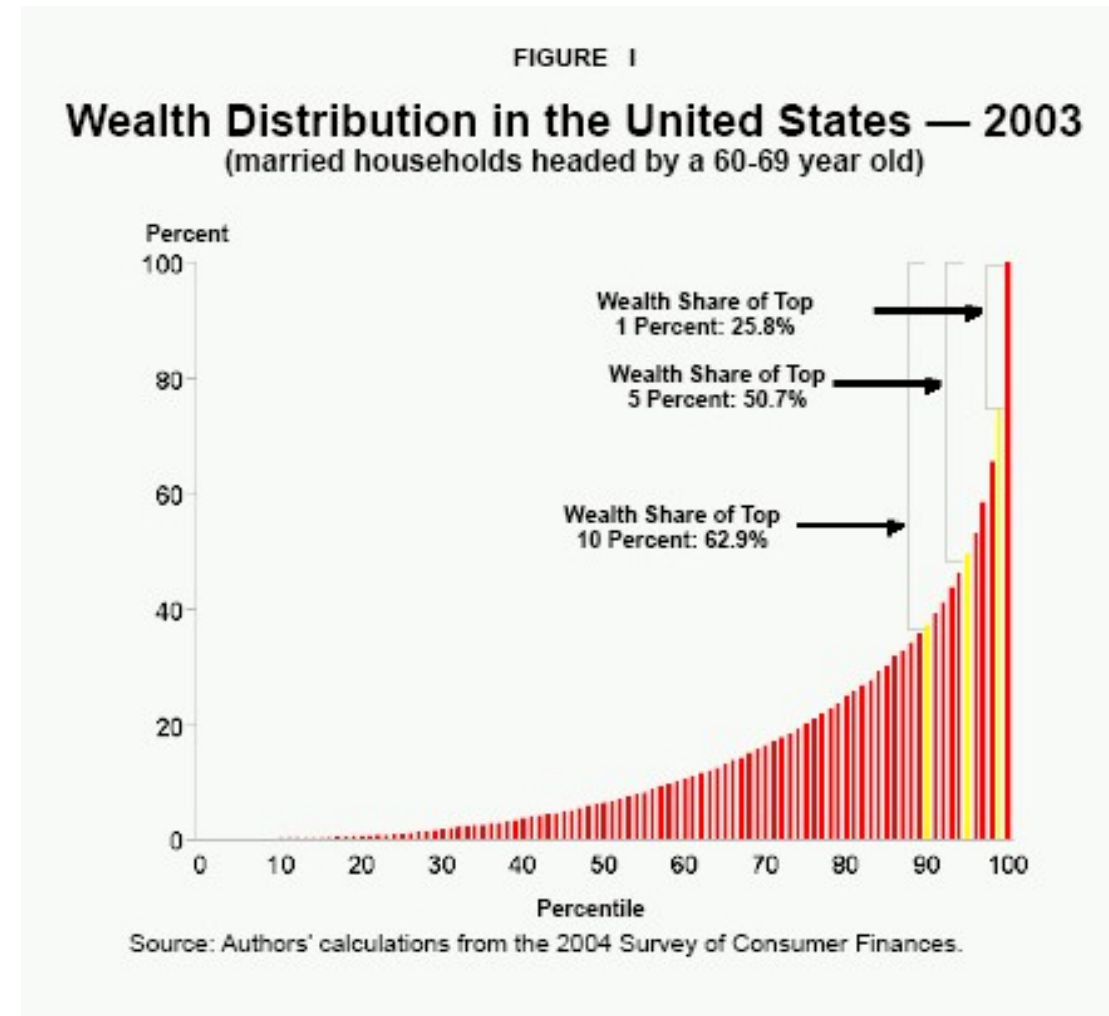


Source: "Subjective Well-Being and Income: Is There Any Evidence of Satiation?", by Betsey Stevenson and Justin Wolfers. NBER Working Paper 18992. April 2013

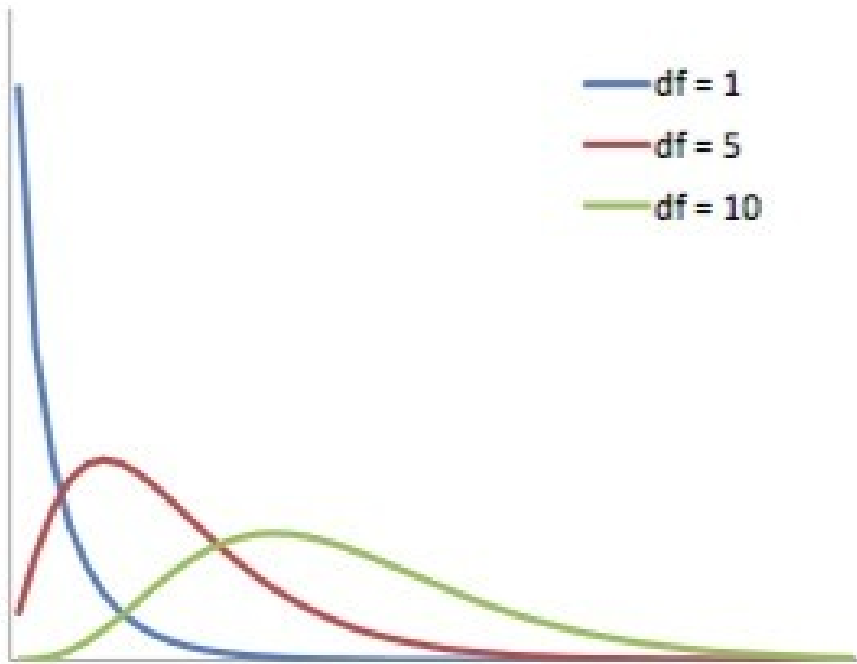
DISTRIBUTIONS WHICH ARE NOT NORMAL



<https://www.deanyeong.com/article/pareto-principle>



CHI SQUARE DISTRIBUTION



If $Z_1, \dots, Z_k$ are independent, standard normal random variables, then the sum of their squares,

$$Q = \sum_{i=1}^k Z_i^2,$$

is distributed according to the chi-squared distribution with k degrees of freedom. This is usually denoted as

$$Q \sim \chi^2(k) \text{ or } Q \sim \chi_k^2.$$

The chi-squared distribution has one parameter: a positive integer k that specifies the number of degrees of freedom (the number of random variables being summed, Z_i s).

Source : Wikipedia
https://en.wikipedia.org/wiki/Chi-squared_distribution

DETERMINING NORMAL PROBABILITIES

When value do not fall directly on σ landmarks:

1. **State** the problem
2. **Standardize** the value(s) (z score)
3. **Sketch**, label, and shade the curve
4. Use **Tables of Normal Distribution (now a days- just use soft wares)**

Z- SCORES

Z-scores Also known as standardized scores, z-scores tell us how many standard deviations a particular case falls from the mean. Cases above the mean have positive z-scores and cases below the mean have negative z-scores.

$$Z = \frac{x - \mu}{\sigma}$$

- z is equal to the raw score that is to be standardized
- μ is equal to the population mean
- σ is equal to the standard deviation of the population

CALCULATING Z SCORE

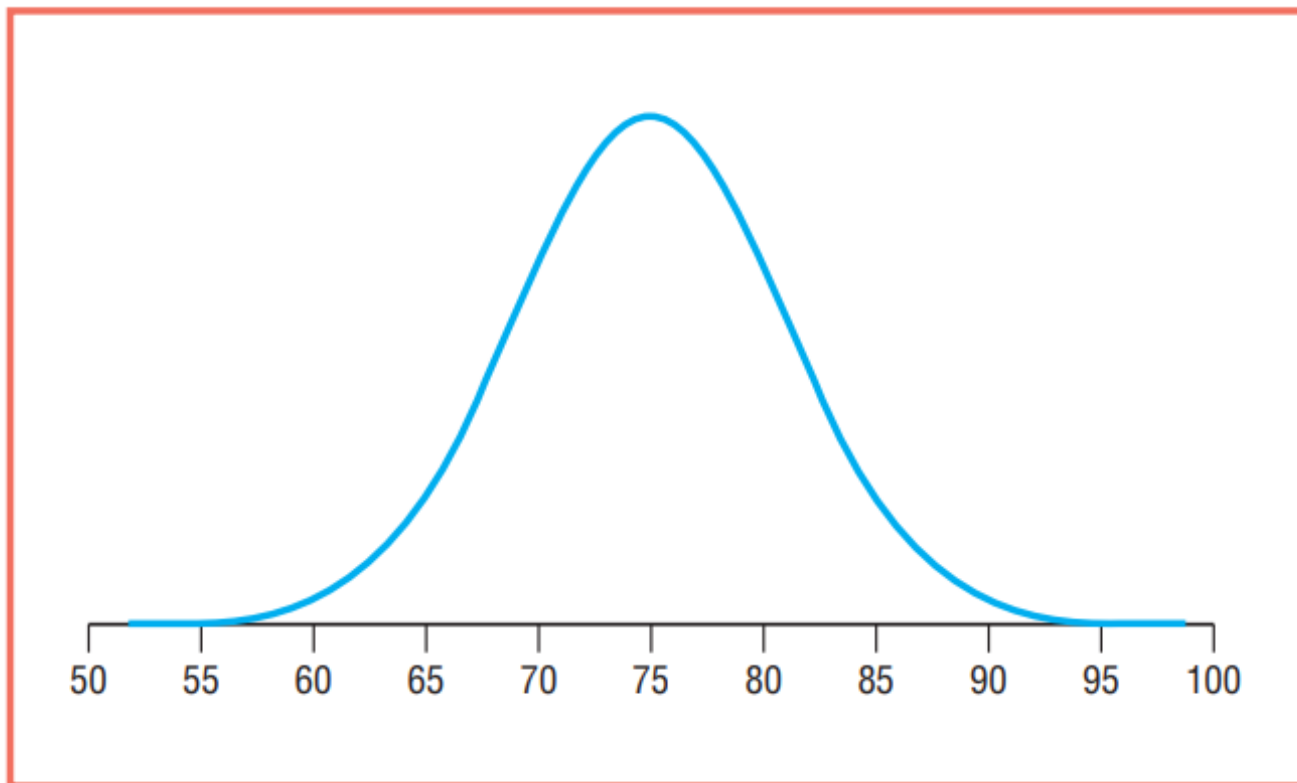


Figure 3 Test Scores

$$\mu = 75, \sigma = 6$$

Student score

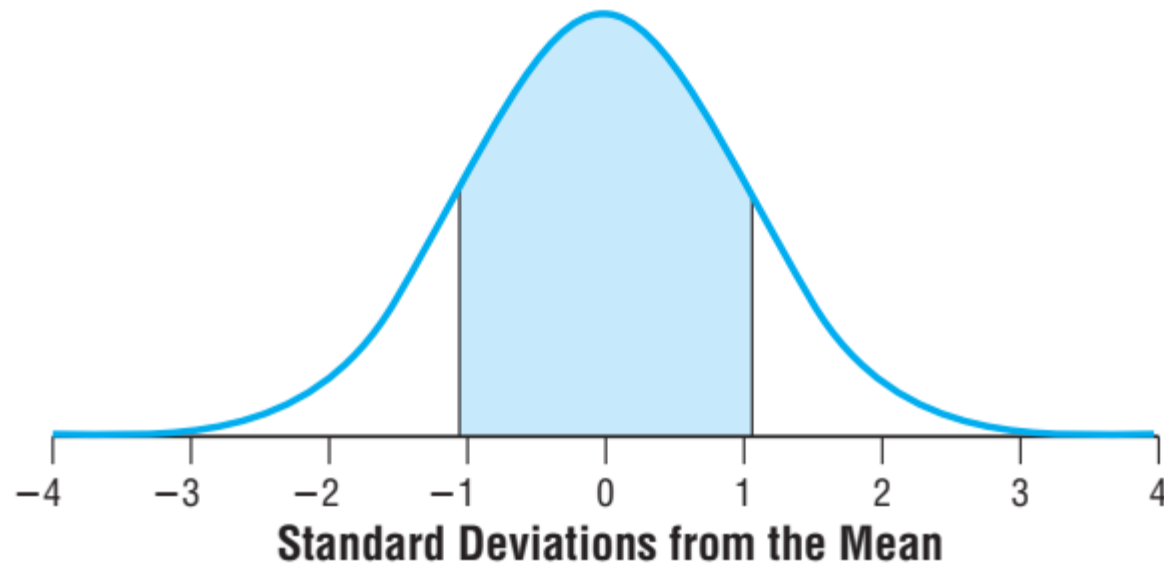
A=81; B=69

$$\text{Student A: } z = \frac{x - \mu}{\sigma} = \frac{81 - 75}{6} = \frac{6}{6} = 1$$

$$\text{Student B: } z = \frac{x - \mu}{\sigma} = \frac{69 - 75}{6} = \frac{-6}{6} = -1$$

Z FOLLOWS STANDARD NORMAL DISTRIBUTIONS

Figure 2 Normal Curve



STEP 1: STATE THE PROBLEM

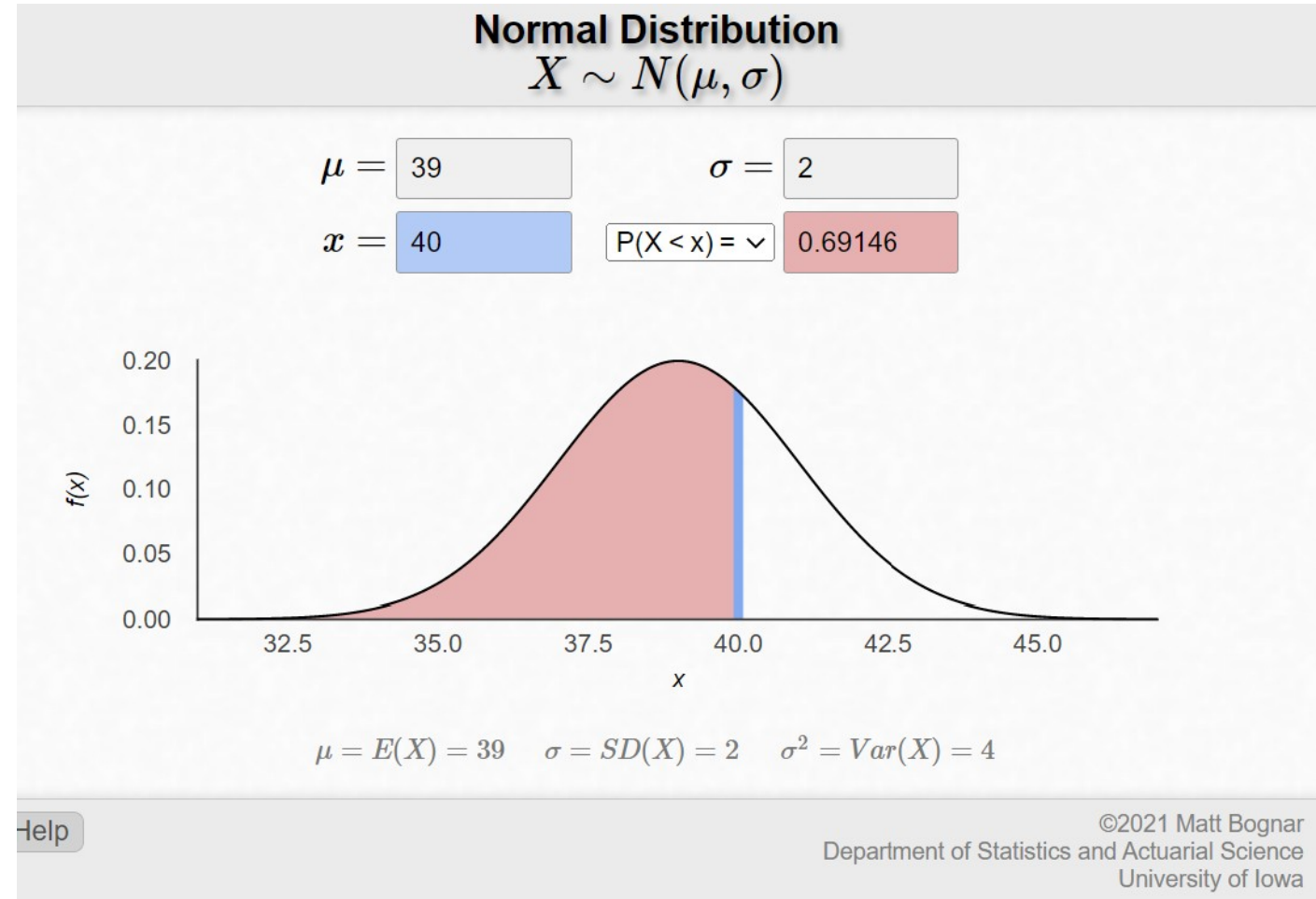
Pregnancy and Gestation period

What percentage of gestations are less than 40 weeks?

Let $X \equiv$ gestational length

We know from prior research:
 $X \sim N(39, 2)$ weeks

$\Pr(X \leq 40) = ?$

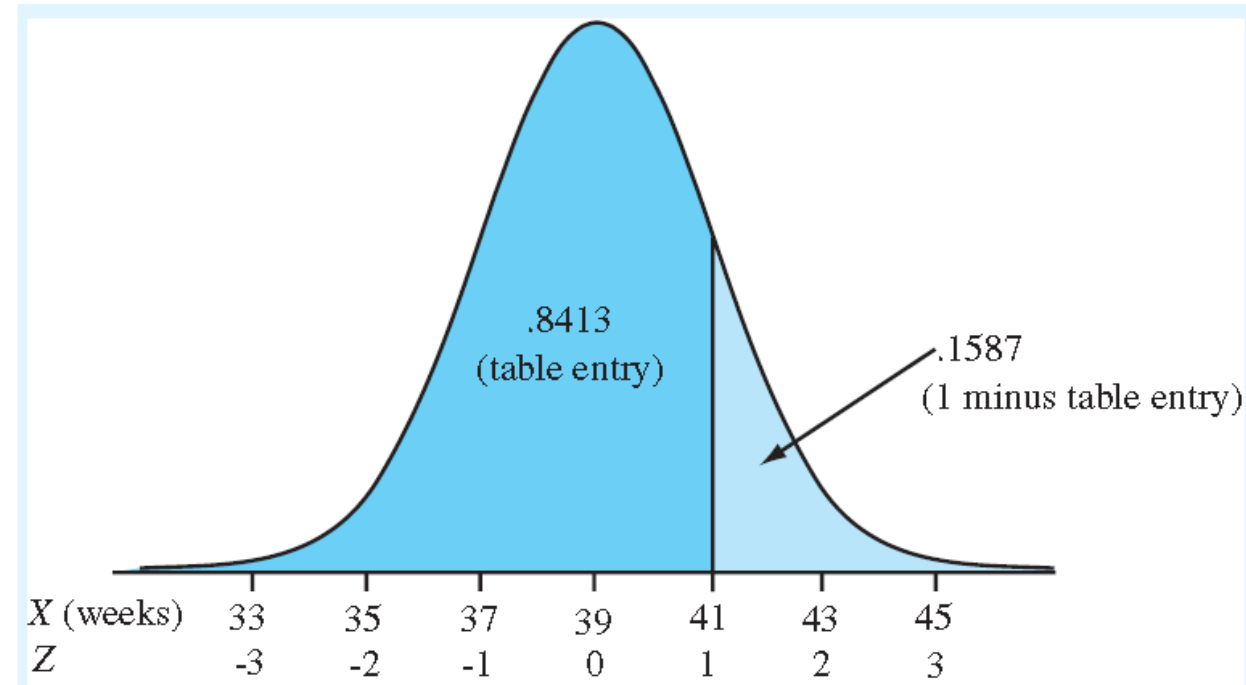


STEP 2: STANDARDIZE

Standard Normal variable $\equiv$
“Z” $\equiv$ a Normal random variable
with $\mu = 0$ and $\sigma = 1$

$$Z \sim N(0,1)$$

Use **Table B** to look up
cumulative probabilities for Z



STEP 2 (CONT.)

Turn value into z score:

$$z = \frac{x - \mu}{\sigma}$$

z-score = no. of σ -units above (positive z) or below (negative z) distribution mean μ

For example, the value 40 from $X \sim N(39, 2)$ has

$$z = \frac{40 - 39}{2} = 0.5$$

$z=0.5$

$x=40$

$P(Z \leq z) = 0.6915$

5

$P(X \leq x) = 0.6915$

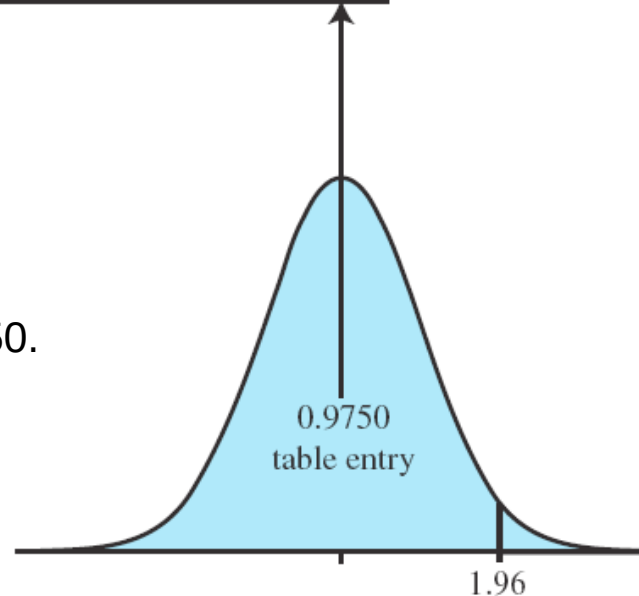
5

z tenths	hundredths									
	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239	.5279	.5319	.5359
0.1	.5398	.5438	.5478	.5517	.5557	.5596	.5636	.5675	.5714	.5753
0.2	.5793	.5832	.5871	.5910	.5948	.5987	.6026	.6064	.6103	.6141
0.3	.6179	.6217	.6255	.6293	.6331	.6368	.6406	.6443	.6480	.6517
0.4	.6554	.6591	.6628	.6664	.6700	.6736	.6772	.6808	.6844	.6879
0.5	.6915	.6950	.6985	.7019	.7054	.7088	.7123	.7157	.7190	.7224
0.6	.7257	.7291	.7324	.7357	.7389	.7422	.7454	.7486	.7517	.7549
0.7	.7580	.7611	.7642	.7673	.7704	.7734	.7764	.7794	.7823	.7852
0.8	.7881	.7910	.7939	.7967	.7995	.8023	.8051	.8078	.8106	.8133
0.9	.8159	.8186	.8212	.8238	.8264	.8289	.8315	.8340	.8365	.8389
1.0	.8413	.8438	.8461	.8485	.8508	.8531	.8554	.8577	.8599	.8621
1.1	.8643	.8665	.8686	.8708	.8729	.8749	.8770	.8790	.8810	.8830
1.2	.8849	.8869	.8888	.8907	.8925	.8944	.8962	.8980	.8997	.9015
1.3	.9032	.9049	.9066	.9082	.9099	.9115	.9131	.9147	.9162	.9177
1.4	.9192	.9207	.9222	.9236	.9251	.9265	.9279	.9292	.9306	.9319
1.5	.9332	.9345	.9357	.9370	.9382	.9394	.9406	.9418	.9429	.9441
1.6	.9452	.9463	.9474	.9484	.9495	.9505	.9515	.9525	.9535	.9545
1.7	.9554	.9564	.9573	.9582	.9591	.9599	.9608	.9616	.9625	.9633
1.8	.9641	.9649	.9656	.9664	.9671	.9678	.9686	.9693	.9699	.9706
1.9	.9713	.9719	.9726	.9732	.9738	.9744	.9750	.9756	.9761	.9767

From Tables (manual approach)

Example:

A z variable of 1.96 has *cumulative* probability 0.9750.

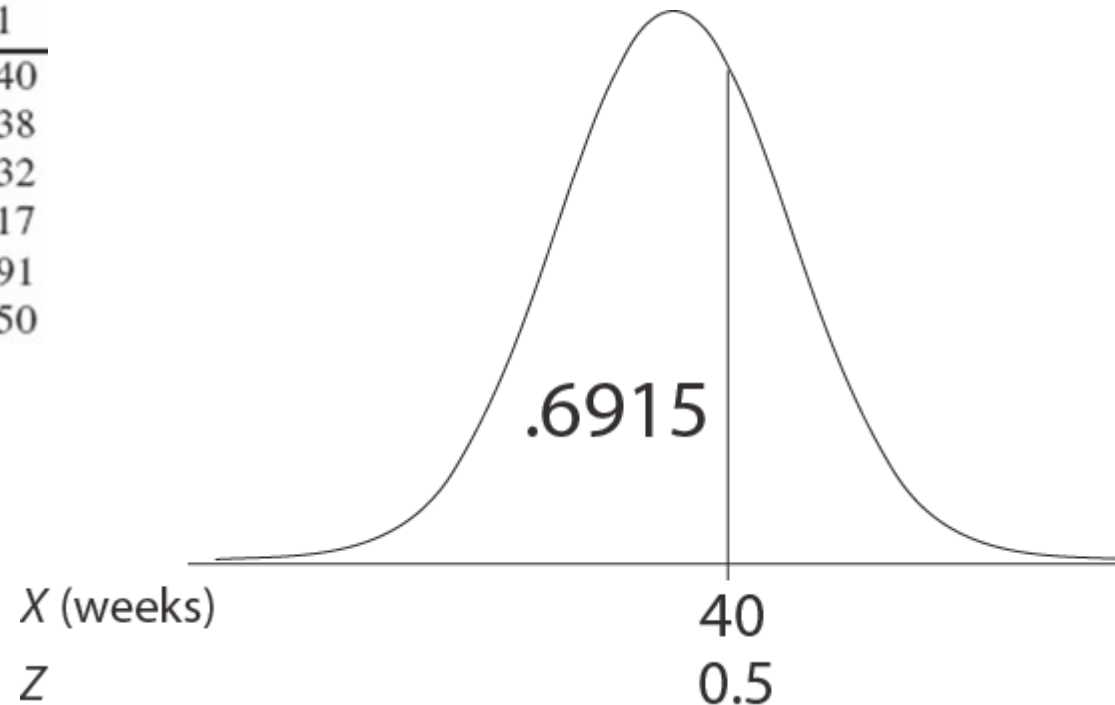


STEPS 3 & 4: SKETCH & TABLE B

3. Sketch

4. Use **Table B** to lookup $\Pr(Z \leq 0.5) = 0.6915$

z tenths	.00	.01
0.0	.5000	.5040
0.1	.5398	.5438
0.2	.5793	.5832
0.3	.6179	.6217
0.4	.6554	.6591
0.5	<u>.6915</u>	.6950

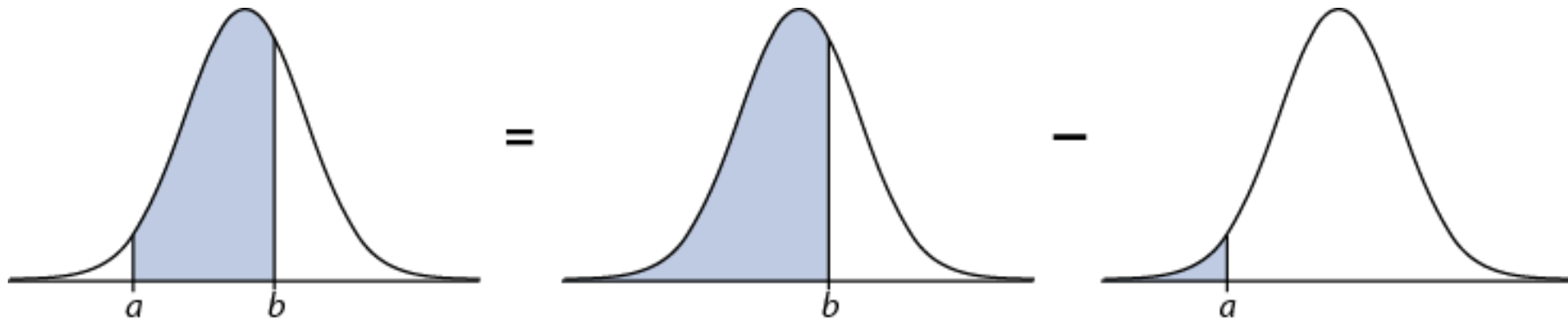


PROBABILITIES BETWEEN POINTS

a represents a lower boundary

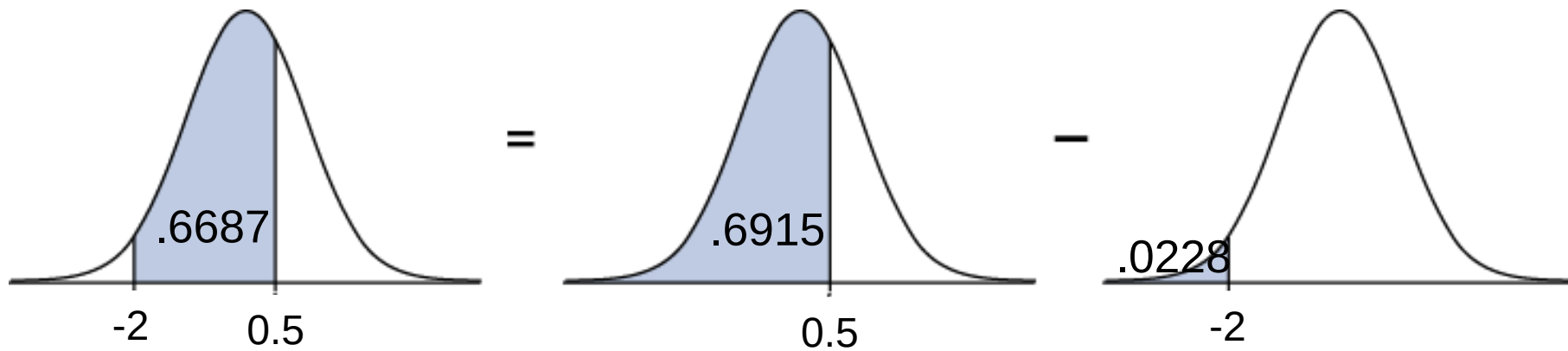
b represents an upper boundary

$$\Pr(a \leq Z \leq b) = \Pr(Z \leq b) - \Pr(Z \leq a)$$



BETWEEN TWO POINTS

$$\begin{aligned}\Pr(-2 \leq Z \leq 0.5) &= \Pr(Z \leq 0.5) - \Pr(Z \leq -2) \\ .6687 &= .6915 - .0228\end{aligned}$$



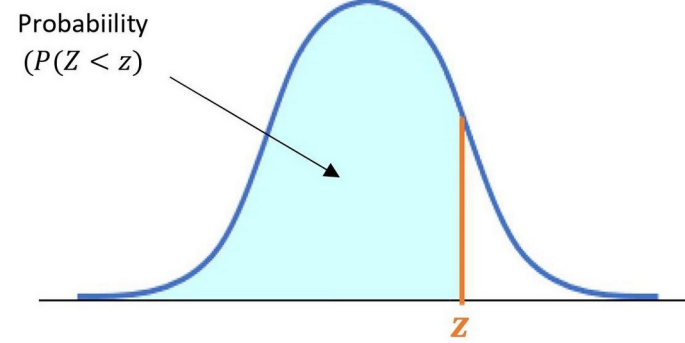
VALUES CORRESPONDING TO NORMAL PROBABILITIES

1. **State** the problem
2. Find **Z-score corresponding to** percentile (Table B)
2. **Sketch**
4. **Unstandardize:**

$$x = \mu + z_p \sigma$$

Z PERCENTILES

- $z_p \equiv$ the Normal z variable with cumulative probability p
- Use **Normal Distribution Table** to look up the value of z_p
- Look *inside* the table for the closest cumulative probability entry
- Trace the z score to row and column

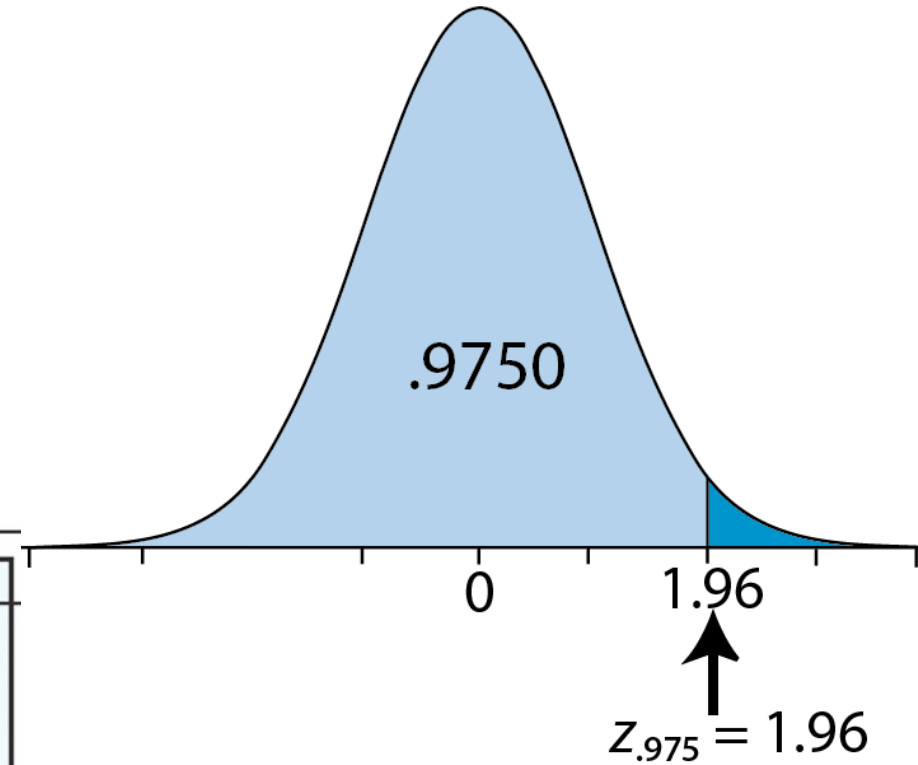


z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5754
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7258	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7518	0.7549
0.7	0.7580	0.7612	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7996	0.8023	0.8051	0.8079	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9430	0.9441
1.6	0.9452	0.9463	0.9474	0.9485	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9700	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9762	0.9767
2.0	0.9773	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9865	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936

e.g., What is the 97.5th percentile on the Standard Normal curve?

$$z_{.975} = 1.96$$

z tenths	hundredths					
	.00	.01	.02	.03	.04	.05
0.0	.5000	.5040	.5080	.5120	.5160	.5199
0.1	.5398	.5438	.5478	.5517	.5557	.5596
0.2	.5793	.5832	.5871	.5910	.5948	.5987
0.3	.6179	.6217	.6255	.6293	.6331	.6368
0.4	.6554	.6591	.6628	.6664	.6700	.6736
0.5	.6915	.6950	.6985	.7019	.7054	.7088
0.6	.7257	.7291	.7324	.7357	.7389	.7422
0.7	.7580	.7611	.7642	.7673	.7704	.7734
0.8	.7881	.7910	.7939	.7967	.7995	.8023
0.9	.8159	.8186	.8212	.8238	.8264	.8289
1.0	.8413	.8438	.8461	.8485	.8508	.8531
1.1	.8643	.8665	.8686	.8708	.8729	.8749
1.2	.8849	.8869	.8888	.8907	.8925	.8944
1.3	.9032	.9049	.9066	.9082	.9099	.9115
1.4	.9192	.9207	.9222	.9236	.9251	.9265
1.5	.9332	.9345	.9357	.9370	.9382	.9394
1.6	.9452	.9463	.9474	.9484	.9495	.9505
1.7	.9554	.9564	.9573	.9582	.9591	.9599
1.8	.9641	.9649	.9656	.9664	.9671	.9678
1.9	.9713	.9719	.9726	.9732	.9738	.9744



Notation: Let z_p represents the z score with cumulative probability p , e.g., $z_{.975} = 1.96$

STEP 1: STATE PROBLEM

Question: *What gestational length is smaller than 97.5% of gestations?*

Let X represent gestations length

We know from prior research that
 $X \sim N(39, 2)$

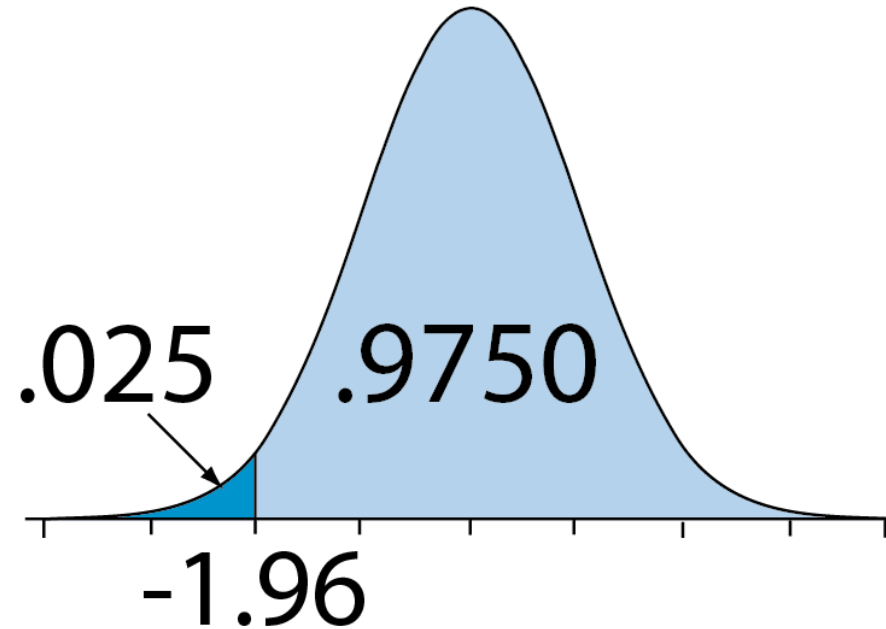
A value that is smaller than .975 of gestations has a cumulative probability of .025

STEP 2 (Z PERCENTILE)

Less than 97.5%
(right tail) =
greater than 2.5%
(left tail)

z lookup:

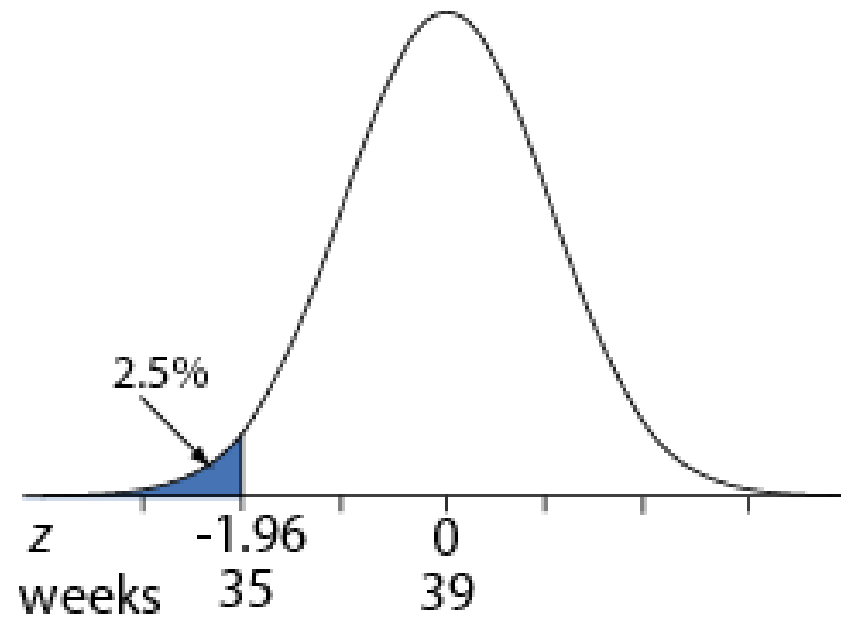
$$Z_{.025} = -1.96$$



z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
-1.9	.0287	.0281	.0274	.0268	.0262	.0256	.0250	.0244	.0239	.0233

UNSTANDARDIZE AND SKETCH

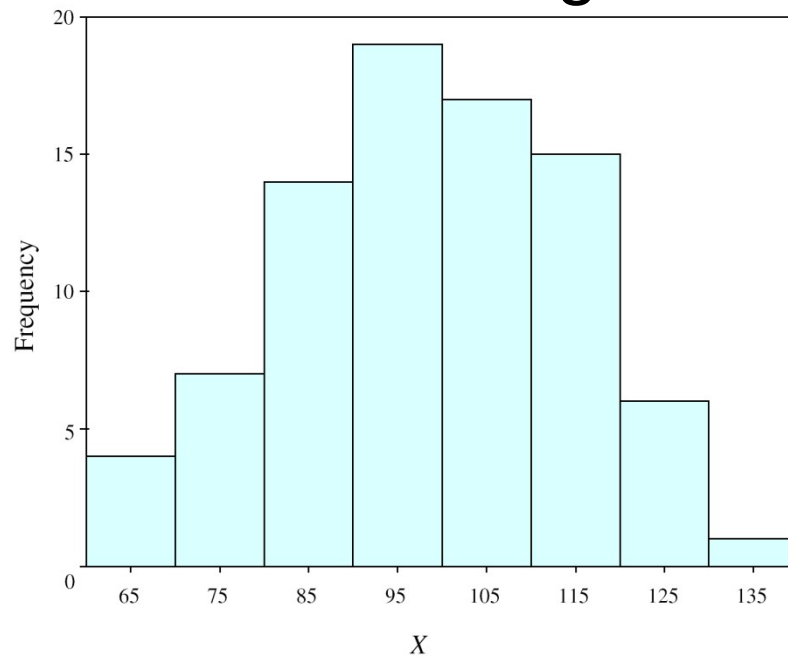
$$x = \mu + z_p \sigma = 39 + (-1.96)(2) \approx 35$$



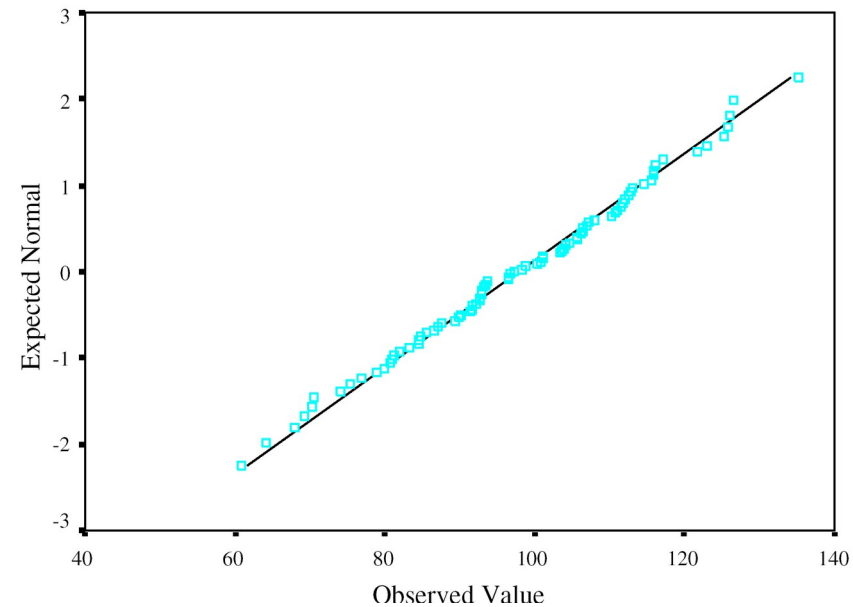
The 2.5th percentile is 35 weeks

ASSESSING DEPARTURES FROM NORMALITY

Approximately Normal histogram

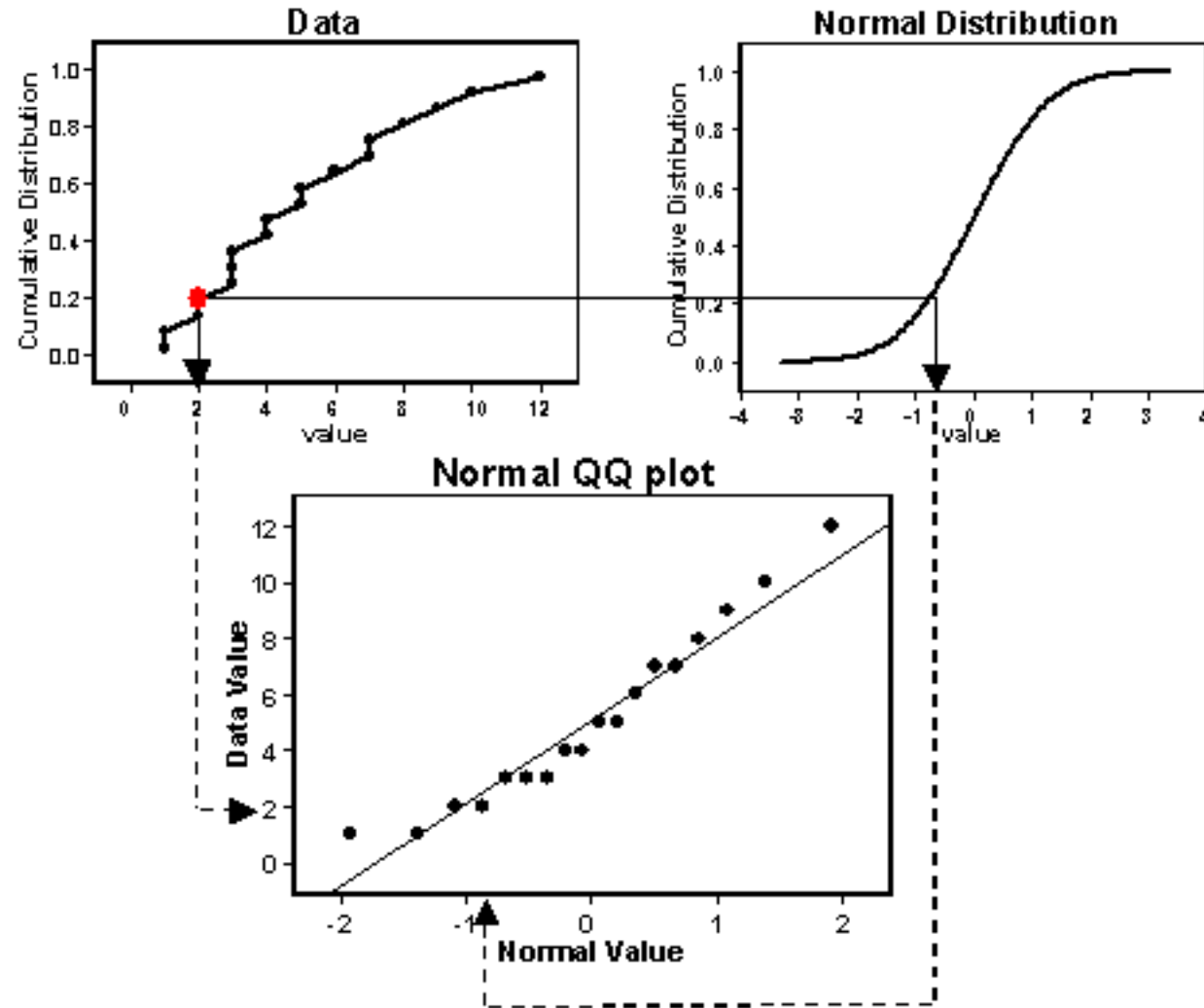


Same distribution on Normal “O-O” Plot



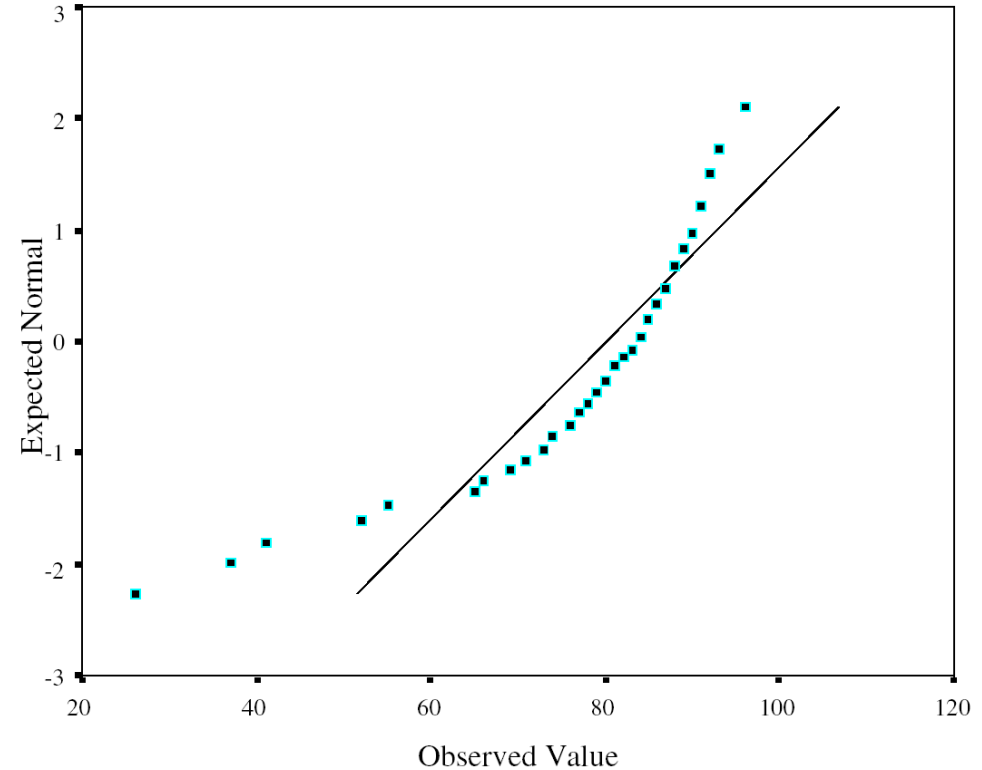
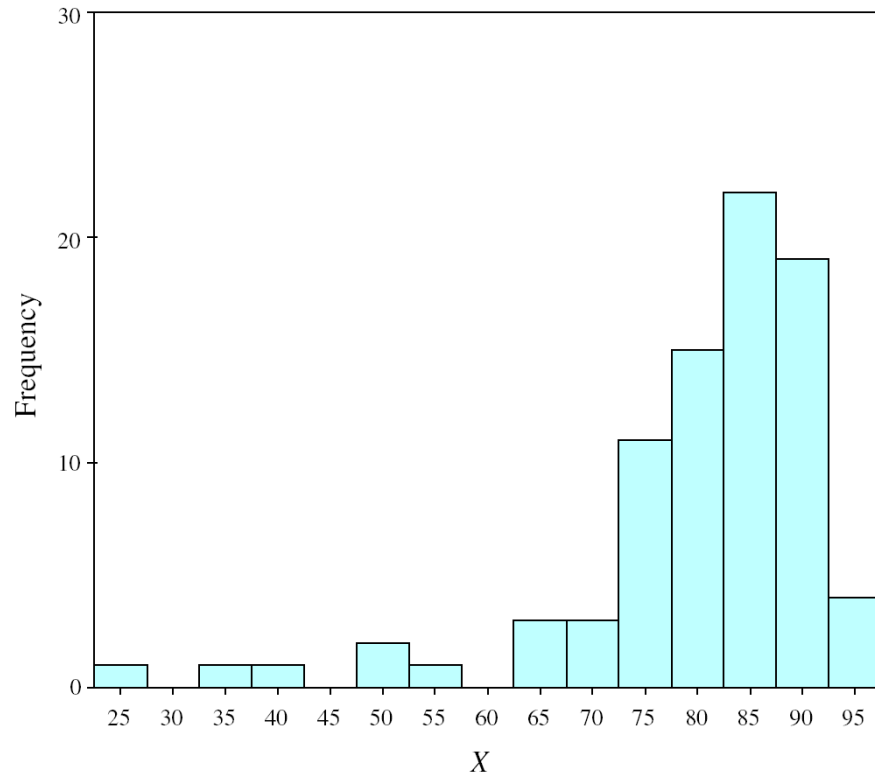
Normal distributions adhere to diagonal line on Q-Q plot

Q-Q PLO



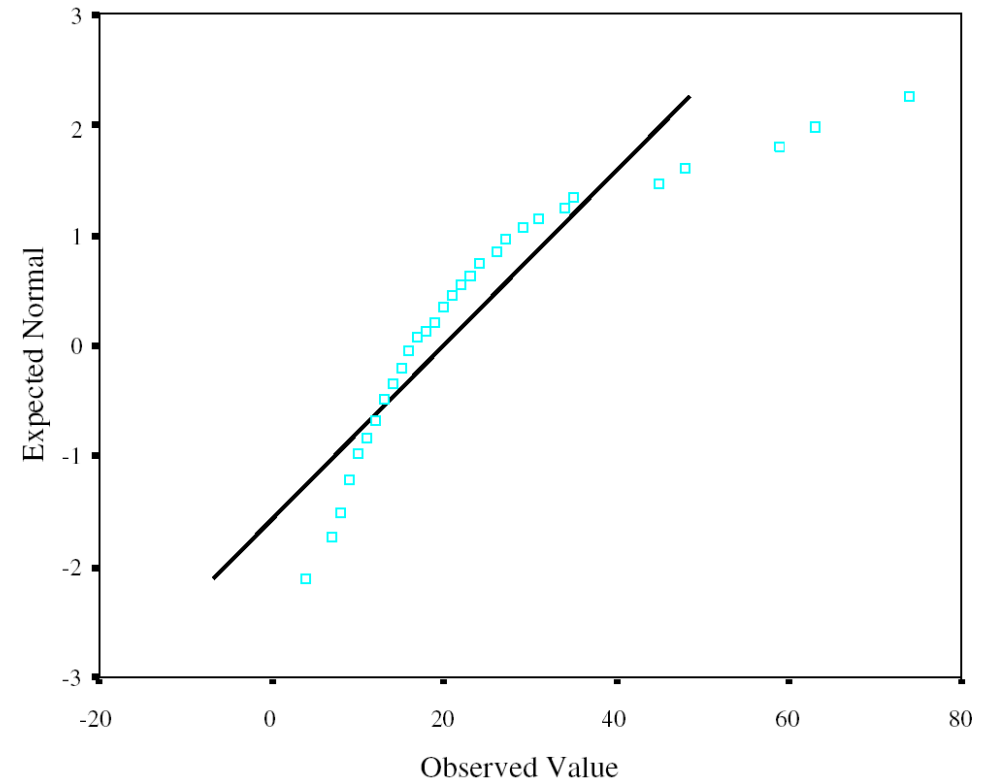
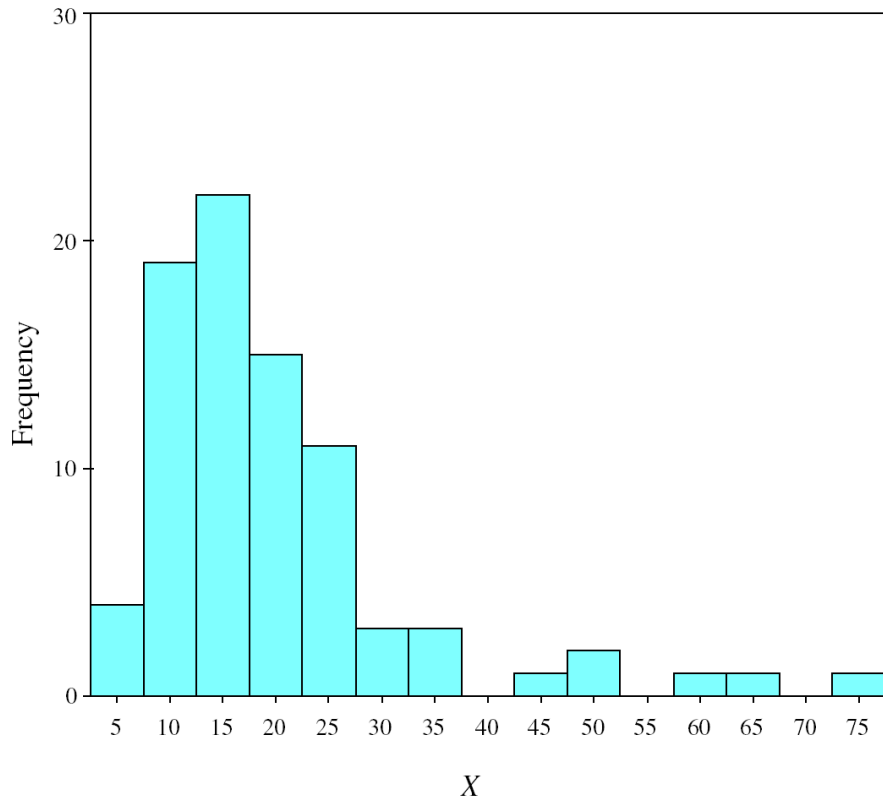
<https://desktop.arcgis.com/en/arcmap/latest/extensions/geostatistical-analyst/normal-qq-plot-and-general-qq-plot.htm>

NEGATIVE SKEW



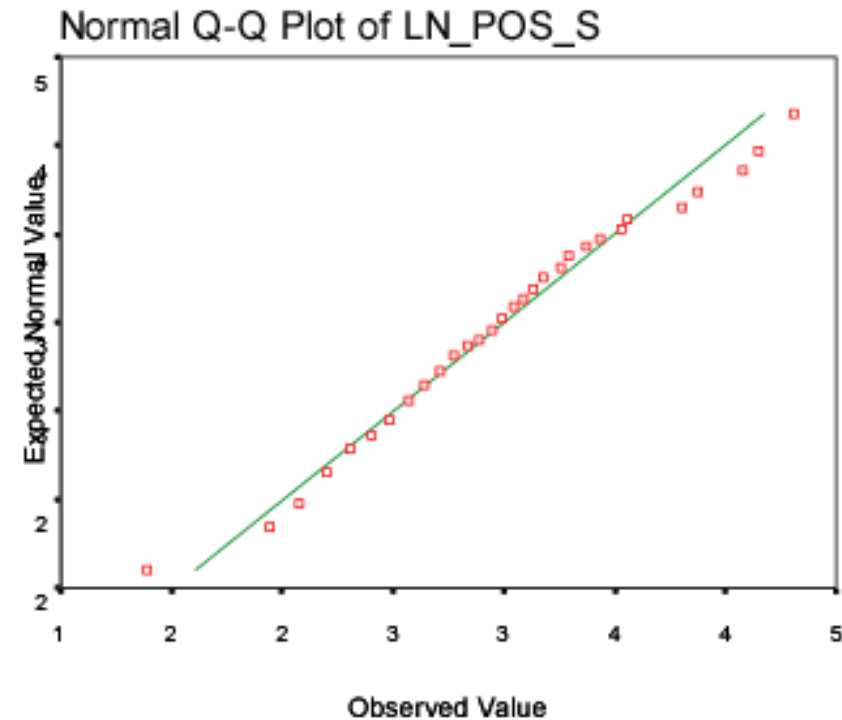
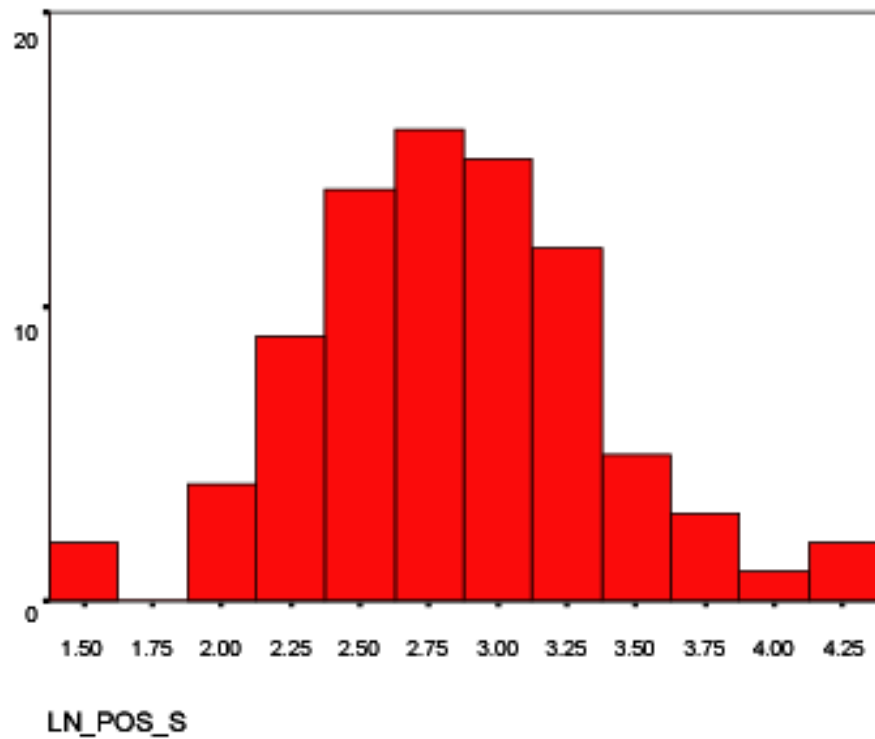
Negative skew shows upward curve on Q-Q plot

POSITIVE SKEW



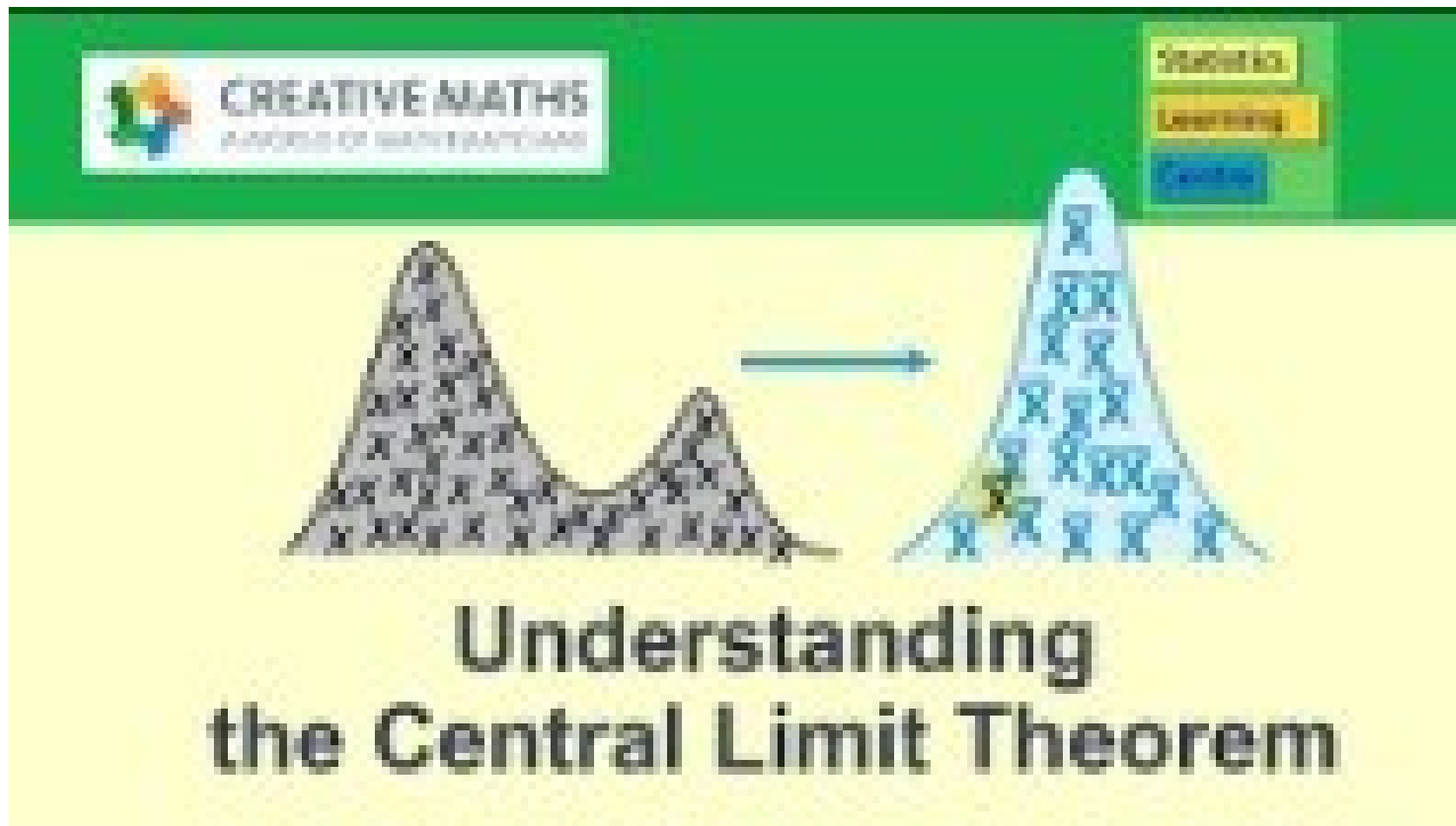
Positive skew shows downward curve on Q-Q plot

SAME DATA AS PRIOR SLIDE WITH LOGARITHMIC TRANSFORMATION



The log transform Normalize the skew

CENTRAL LIMIT THEOREM



https://www.youtube.com/watch?v=_YOr_yYPyTM